

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Autonomous solver submission

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Abstract

This report documents an original C++17/MPI cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, applied to the eight required NACA0012 and cylinder benchmark cases. The solver reads CGNS meshes, partitions the cell graph with METIS, exchanges neighbor halo data with MPI, and marches in pseudo-time with a simplified-Jacobian LU-SGS implicit scheme, Venkatakrishnan-limited Green–Gauss second-order reconstruction, and the case-supplied CFL continuation schedule; the cylinder Re 200 case uses a true BDF2 dual-time transient loop whose inner nonlinear solve is a matrix-free Newton–GMRES iteration with frozen physical-time histories. All eight production runs satisfy the benchmark output contract and validation gate described in Section 8.

1 Introduction

This benchmark requires a 2-D unstructured finite-volume compressible-flow solver for the eight cases supplied in the benchmark input repository, with final result packages following the benchmark output contract. Section 2 defines the equations; Section 3 the meshes and inputs; Sections 4–7 the spatial scheme, boundary conditions, time integration, and MPI design; Sections 9–11 present the results, parallel validation, and limitations.

2 Governing Equations and Nondimensionalization

The conservative state is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \quad (1)$$

and the 2-D compressible Navier–Stokes equations read

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad (2)$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \quad (3)$$

The gas is calorically perfect,

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}, \quad (4)$$

with case-supplied γ , R , and Prandtl number. Viscous cases use the Newtonian stress tensor

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \quad (5)$$

Fourier heat flux $\mathbf{q} = -k\nabla T$ with $k = \mu c_p / \text{Pr}$, and a constant viscosity matched to the case Reynolds number, $\mu = \rho_\infty U_\infty L_{\text{ref}} / \text{Re}$. All quantities are used directly in the nondimensional units of the case files ($\rho_\infty = 1$, $U_\infty = 1$, $p_\infty = 1/(\gamma M_\infty^2)$). Force coefficients use the dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$ and the case reference area/length, $C_F = F/(q_\infty A_{\text{ref}})$.

3 Meshes, Boundary Tags, and Case Inputs

Both CGNS meshes are unstructured mixed-cell meshes. The NACA0012 mesh ($-80 \leq x, y \leq 80$) has a single zone with 10,752 triangles and 10,064 quads (20,816 cells), an 80-face outer farfield boundary (family **bc-2**) and a 404-face airfoil wall (**bc-4**). The cylinder mesh ($-200 \leq x, y \leq 200$) has two zones joined by five 1-to-1 abutting connectivity patches (vertex pairs merged through union-find): an inner O-block with the 100-face **WALL** and an outer block with the 20-face **FAR** farfield (10,185 cells total). Cell volumes use the shoelace formula, centroids the vertex average, and each deduplicated edge carries an area (length), midpoint, and outward unit normal oriented from the face's left cell; boundary families are mapped to farfield, slip-wall, or no-slip-adiabatic-wall conditions through the case JSON mapping only (no hard-coded names). The case inputs (mesh, mode, CFL continuation, stopping horizon) are summarized in Table 5.

4 Spatial Discretization

The cell-centered finite-volume residual is

$$\mathbf{R}_i = \frac{1}{V_i} \sum_{f \in i} [\mathbf{F}^i(\mathbf{U}_f) - \mathbf{F}^v(\mathbf{U}_f, \nabla \mathbf{U}_f)] \cdot \hat{\mathbf{n}}_f A_f, \quad (6)$$

summed over every face of cell i with outward unit normal $\hat{\mathbf{n}}_f$.

4.1 Second-Order Reconstruction

Primitive gradients ($\nabla \rho, \nabla u, \nabla v, \nabla p$) are computed with the Green–Gauss formula $\nabla q_i = \frac{1}{V_i} \sum_f \frac{1}{2}(q_i + q_j) \hat{\mathbf{n}}_f A_f$ (boundary faces use the boundary state). Face states are the piecewise-linear reconstruction $q_f = q_i + \phi_i \nabla q_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$ for both sides of every interior face. Reconstruction and limiting are active in all production runs; the only first-order per-face fallback is the positivity trigger described below, and its activation count is recorded in each case's **metadata.json**. (The **CFD_RECON_SCALE=0.5** damping described next is a global reconstruction scaling, not a per-face fallback.)

Three steady cases (NACA M=0.15 inviscid, NACA M=0.8 laminar, and the cylinder Re 20 case) additionally damp the reconstruction with **CFD_RECON_SCALE=0.5**, i.e. the limited gradient entering the face reconstruction is scaled by 0.5. This is a documented first-order blend of the *inviscid* reconstruction only: wall skin friction is evaluated from the unscaled cell-centered gradient with the cell-center ghost anchor, so the reported skin friction is unaffected, while the interior-face viscous gradient correction uses the same scaled face states as the inviscid flux and therefore inherits the damping on these three cases. It suppresses the marginal trailing-edge and near-wake odd-even limit cycle that the full second-order operator otherwise sustains on these three cases. The remaining production cases use the full reconstruction (**CFD_RECON_SCALE** = 1).

4.2 Limiter and Positivity Control

The production runs use the per-variable Venkatakrishnan limiter $\phi_i = \min_f \Phi(\Delta_f^+, \Delta_f^-)$ with smoothing parameter $K = 0.3$, which relaxes the Barth–Jespersen clipping near smooth extrema while retaining monotonicity control across shocks (a Barth–Jespersen variant remains available through `CFD.LIMITER=barth`). The limiter is applied independently to ρ , u , v , and p . If a reconstructed face state loses density or pressure positivity it falls back to the cell-centered (first-order) state for that face, and the number of such fallback faces is recorded in each case’s metadata; after every update the conservative state is clamped so that $\rho > 10^{-10}\rho_\infty$ and $p > 10^{-10}p_\infty$.

4.3 Inviscid and Viscous Fluxes

The inviscid face flux is the Rusanov (local Lax–Friedrichs) flux, $\mathbf{F}_f = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2}\lambda_{\max}(\mathbf{U}_R - \mathbf{U}_L)$ with $\lambda_{\max} = \max(|\mathbf{v}_L \cdot \hat{\mathbf{n}}| + a_L, |\mathbf{v}_R \cdot \hat{\mathbf{n}}| + a_R)$, for the steady inviscid and supersonic production runs, where the flow and acoustic scales are comparable. The low-Mach viscous production runs (NACA M=0.15/M=0.8 laminar, cylinder Re 20) and the cylinder Re 200 transient run use the Roe flux with the Harten–Hyman entropy fix, because Rusanov’s acoustic-scale dissipation at $M_\infty \leq 0.15$ acts as an artificial viscosity two orders of magnitude above the physical μ (Section 11 documents the comparison). Viscous fluxes use the face-averaged primitive state and the face gradient formed from the averaged cell gradients plus the directional correction $\nabla q_f = \overline{\nabla q} + [(q_R - q_L)/d - \overline{\nabla q} \cdot \hat{\mathbf{s}}] \hat{\mathbf{s}}$, $d = |\mathbf{x}_R - \mathbf{x}_L|$.

5 Boundary Conditions

Farfield. The characteristic-variable farfield state (Riemann invariants $R^\pm = v_n \pm 2a/(\gamma - 1)$, entropy and tangential velocity from the interior on outflow faces and from the freestream on inflow faces) is evaluated from the interior state, blended weakly toward the freestream, and applied as an upwind Rusanov-type face flux against the interior state so boundary-face modes stay damped; supersonic faces use the interior (outflow) or freestream (inflow) state. The viscous farfield flux uses the interior gradient.

Inviscid slip wall. The boundary state mirrors the normal velocity ($\mathbf{v}_w = \mathbf{v}_i - 2(\mathbf{v}_i \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$) and the wall flux is the upwind flux between the reconstructed interior state and the mirrored state; its dissipative term vanishes as the slip condition $v_n \rightarrow 0$ is approached, so normal mass flux is zero and tangential velocity is preserved. A pure pressure flux is available through `CFD.PRESSURE_WALL=1`.

Viscous no-slip adiabatic wall. The boundary state is the mirrored state $(\rho, -u, -v, p)$ (wall velocity exactly zero) and the wall flux is the pressure flux plus the viscous flux evaluated with the mirror-corrected wall-normal gradient and the wall state ($u = v = 0$, $T_w = T_i$, adiabatic normal heat flux zero). Surface output reports these boundary values, not adjacent cell-center values; skin friction is the tangential part of the wall traction, kept separate from the pressure force.

6 Implicit and Transient Time Integration

6.1 Steady Cases

Steady cases use local pseudo-time stepping $\Delta t_i = \text{CFL } V_i / \left(\sum_f (|\mathbf{v}_f \cdot \hat{\mathbf{n}}_f| + a_f) A_f + \sum_f 2 \max(\frac{4}{3}, \gamma) \frac{\mu}{\rho_i \text{Pr}} \frac{A_f^2}{V_i} \right)$ with the case-supplied CFL ramp ($\text{CFL}_0 \rightarrow \text{CFL}_{\max}$ over `pseudo_cfl_ramp_steps`, `CFD_NO_WATCHDOG=1`), and a simplified-Jacobian LU-SGS inner solve per nonlinear step: each inner iteration re-evaluates the full nonlinear residual and applies one forward and one backward sweep, $(D + L)D^{-1}(D + U)\Delta \mathbf{U} = -\mathbf{R}$, with $D_i = V_i/\Delta t_i + \sum_f \lambda_f + \sum_f \nu_f$ and the off-diagonal blocks replaced by their scalar spectral radii. Between sweeps the reconstruction gradients are frozen once the inner residual falls near its target and are refreshed (refreeze phase) until the inner target is met, which keeps the implicit system consistent while the limiter remains active. Inner iterations obey the case `min_inner_iterations`, `max_inner_iterations`, and `inner_residual_reduction_target`. Global residual and force norms use MPI reductions. Convergence is declared when the global L2 residual drops by the case target orders and the drag history plateaus (relative variation below 5×10^{-5} over the trailing 500 steps). The per-step inner-iteration count is recorded in the `inner_iter` column of each `residuals.csv`; the submitted steady production runs consumed the full case-supplied inner budget on every step (50–100 iterations depending on the case), so the steady inner solve is a fixed-budget relaxation rather than an early-converging one.

6.2 Cylinder Re 200: BDF2 Dual-Time Stepping

The Re 200 run uses the supplied production controls ($\Delta t = 0.01$, $t_f = 300$, 30,000 physical steps, pseudo-CFL fixed at 1.0). The inviscid face flux is the Roe flux with the Harten–Hyman entropy fix: at the benchmark Mach number $M_\infty = 0.1$ the Rusanov/LLF dissipation acts at the acoustic scale and its numerical viscosity on this mesh exceeds the physical cylinder viscosity by two orders of magnitude, which suppresses the recirculating wake and the vortex-street instability entirely (this was verified with a full 30,000-step Rusanov production run, retained as a diagnostic). Each physical step solves

$$\frac{V_i}{\Delta \tau} \Delta \mathbf{U}^m = -\mathbf{R}_i(\mathbf{U}^m) - \frac{V_i}{\Delta t} \left(\frac{3}{2} \mathbf{U}^m - 2\mathbf{U}^n + \frac{1}{2} \mathbf{U}^{n-1} \right) \quad (7)$$

with a matrix-free Newton–GMRES inner solve (`CFD_GMRES=1`): the Jacobian-vector products use forward finite differences on the frozen dual-time residual, GMRES(30) is right-preconditioned by one LU-SGS sweep, and a backtracking line search guards the Newton step, until the total transient residual (spatial fluxes plus the physical-time source) drops below 10^{-3} of the step-initial value (5 to 1000 iterations). $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen during all inner iterations and updated only after acceptance; the first physical step uses backward Euler for BDF2 startup. The inner-solve statistics (min/mean/max iterations, target misses, converged fraction, final ratio) are recorded in `metadata.json`.

7 MPI Parallelization and METIS Partitioning

Rank 0 reads the mesh once (serial preprocessing), builds the cell-face adjacency graph, partitions the cell graph with METIS (`METIS_PartGraphKway`, cut objective), and writes one binary partition file per rank; every solver rank then loads only its own owned cells plus the one ghost-cell layer required by the stencil. Solver iterations therefore never replicate the full mesh or the full conservative state. Ghost states and the LU-SGS update guess are exchanged with neighbor ranks through

nonblocking `MPI_Isend/MPI_Irecv` point-to-point messages only (no `Allgather` of states); limited primitive gradients are exchanged in the reverse direction, and residuals/forces use `MPI_Allreduce`. Partition diagnostics (owned/ghost cells, boundary faces, neighbor ranks, send/receive sizes, edge cut, load balance) are written per case; Section 10 compares np=1 and np=8 runs. The per-rank partition diagnostics for the np=8 comparison cases are tabulated in Table 3.

8 Output, Reproducibility, and Sanity Checks

Build, run, and plot commands are listed in `README.md` and the run manifest (`run_manifest.csv`). Every figure is traced to its source CSV or VTU file by `figure_manifest.csv`; the machine-readable physics gate `sanity_checks.json` records positivity of all submitted fields, zero-lift symmetry of the NACA cases, positive cylinder mean drag, Re 200 unsteady lift variation, wall-boundary-value checks, and Mach/pressure figure mapping.

9 Results

Table 1: Run status for every required case.

case	ranks	steps	t_f	wall (s)	resid. orders	status
naca0012_m015_inviscid	8	1091	0	527.7	9.8	converged
naca0012_m080_inviscid	8	961	0	1994.5	4.44	converged
naca0012_m200_inviscid	8	644	0	986.8	3.77	converged
naca0012_m015_laminar_re5000	8	1709	0	3536.5	4.87	converged
naca0012_m080_laminar_re5000	8	1997	0	1027.0	5.93	converged
naca0012_m200_laminar_re5000	8	5587	0	5782.5	3.27	converged
cylinder_m010_laminar_re20	8	969	0	757.8	6.97	converged
cylinder_m010_laminar_re200	8	30000	300	10063.8	0.0	statistically-periodic

Table 2: Final force coefficients for every required case.

case	C_D	C_L	C_{m_z}	$C_{D,p}$	$C_{D,v}$
naca0012_m015_inviscid	0.03439	0.00382	0.00126	0.03439	0.00000
naca0012_m080_inviscid	0.00981	0.00001	0.00027	0.00981	0.00000
naca0012_m200_inviscid	0.09232	0.00058	-0.00002	0.09232	0.00000
naca0012_m015_laminar_re5000	0.01180	0.00034	0.00010	0.00519	0.00660
naca0012_m080_laminar_re5000	0.06957	-0.00743	-0.00180	0.04814	0.02143
naca0012_m200_laminar_re5000	0.10132	0.00025	-0.00010	0.09279	0.00853
cylinder_m010_laminar_re20	2.10341	0.00554	0.00103	1.51827	0.58513
cylinder_m010_laminar_re200	1.14607	0.44380	-0.00038	1.08306	0.06301

9.1 NACA0012 M=0.15 inviscid

Status: **converged**. 1091 pseudo-time steps, residual reduction 9.80 orders (final L2 1.575e-10). Final force coefficients $C_D = 0.03439$, $C_L = 0.00382$, $C_{m_z} = 0.00126$; the final force row corresponds

Table 3: Per-rank partition diagnostics (owned/ghost cells, neighbor ranks, send/receive volumes) for the two np=8 rank-comparison cases, with the global edge cut and load balance from the case summaries.

case	rank	owned	ghost	neighbors	send	recv
naca0012_m015_inviscid	0	2616	96	2	96	96
naca0012_m015_inviscid	1	2600	129	4	129	129
naca0012_m015_inviscid	2	2602	64	4	64	64
naca0012_m015_inviscid	3	2602	165	6	165	165
naca0012_m015_inviscid	4	2585	86	2	85	86
naca0012_m015_inviscid	5	2602	111	3	113	111
naca0012_m015_inviscid	6	2603	120	4	119	120
naca0012_m015_inviscid	7	2606	103	3	103	103
cylinder_m010_laminar_re20	0	1281	96	3	97	96
cylinder_m010_laminar_re20	1	1274	111	6	112	111
cylinder_m010_laminar_re20	2	1266	121	5	120	121
cylinder_m010_laminar_re20	3	1282	123	5	124	123
cylinder_m010_laminar_re20	4	1266	117	4	115	117
cylinder_m010_laminar_re20	5	1274	106	3	107	106
cylinder_m010_laminar_re20	6	1273	120	4	119	120
cylinder_m010_laminar_re20	7	1269	113	4	113	113
naca0012_m015_inviscid (summary)	–	–	–	edge cut 443	–	load balance 1.0054
cylinder_m010_laminar_re20 (summary)	–	–	–	edge cut 476	–	load balance 1.0070

Table 4: Rank-count consistency: np=1 reruns compared with the np=8 production runs for one NACA and one cylinder case (same production controls, converged force coefficients).

case	np	steps	wall (s)	C_D	C_L	$ \Delta C_D $ vs np=8
naca0012_m015_inviscid	8	1091	527.7	0.03439	0.00382	0.000e+00
naca0012_m015_inviscid	1	1095	1948.0	0.03474	-0.00012	3.491e-04
cylinder_m010_laminar_re20	8	969	757.8	2.10341	0.00554	0.000e+00
cylinder_m010_laminar_re20	1	1104	1944.9	2.12662	0.00570	2.322e-02

Table 5: Supplied case inputs: mesh, physics mode, freestream, Reynolds number, CFL continuation, and stopping horizon.

case	mesh	mode	M_∞	Re	CFL_0	$CFL_{\max}$	steps/ t_f
naca0012_m015_inviscid	NACA0012.H2.cgns	inviscid	0.15	-	1.00	100.00	20000 steps
naca0012_m080_inviscid	NACA0012.H2.cgns	inviscid	0.80	-	1.00	100.00	30000 steps
naca0012_m200_inviscid	NACA0012.H2.cgns	inviscid	2.00	-	0.20	50.00	40000 steps
naca0012_m015_laminar_re5000	NACA0012.H2.cgns	laminar	0.15	5000	1.00	100.00	40000 steps
naca0012_m080_laminar_re5000	NACA0012.H2.cgns	laminar	0.80	5000	0.50	80.00	40000 steps
naca0012_m200_laminar_re5000	NACA0012.H2.cgns	laminar	2.00	5000	0.20	50.00	50000 steps
cylinder_m010_laminar_re20	CylinderB1.cgns	laminar	0.10	20	1.00	100.00	30000 steps
cylinder_m010_laminar_re200	CylinderB1.cgns	laminar	0.10	200	1.00	1.00	$t_f=300$

to the final field/surface state. The evidence figures for this case are 15 (residual history), 16 (force history), 18 (Mach field), 19 (near-body pressure field), 20 (full-domain pressure field), 17 (wall C_p /skin-friction distribution).

9.2 NACA0012 M=0.8 inviscid

Status: **converged**. 961 pseudo-time steps, residual reduction 4.44 orders (final L2 3.620e-05). Final force coefficients $C_D = 0.00981$, $C_L = 0.00001$, $C_{m_z} = 0.00027$; the final force row corresponds to the final field/surface state. The evidence figures for this case are 27 (residual history), 28 (force history), 30 (Mach field), 31 (near-body pressure field), 32 (full-domain pressure field), 29 (wall C_p /skin-friction distribution).

9.3 NACA0012 M=2.0 inviscid

Status: **converged**. 644 pseudo-time steps, residual reduction 3.77 orders (final L2 1.694e-04). Final force coefficients $C_D = 0.09232$, $C_L = 0.00058$, $C_{m_z} = -0.00002$; the final force row corresponds to the final field/surface state. The evidence figures for this case are 39 (residual history), 40 (force history), 42 (Mach field), 43 (near-body pressure field), 44 (full-domain pressure field), 41 (wall C_p /skin-friction distribution).

9.4 NACA0012 M=0.15 laminar Re=5000

Status: **converged**. 1709 pseudo-time steps, residual reduction 4.87 orders (final L2 1.363e-05). Final force coefficients $C_D = 0.01180$, $C_L = 0.00034$, $C_{m_z} = 0.00010$; the final force row corresponds to the final field/surface state. The evidence figures for this case are 21 (residual history), 22 (force history), 24 (Mach field), 25 (near-body pressure field), 26 (full-domain pressure field), 23 (wall C_p /skin-friction distribution).

9.5 NACA0012 M=0.8 laminar Re=5000

Status: **converged**. 1997 pseudo-time steps, residual reduction 5.93 orders (final L2 1.185e-06). Final force coefficients $C_D = 0.06957$, $C_L = -0.00743$, $C_{m_z} = -0.00180$; the final force row corresponds to the final field/surface state. The evidence figures for this case are 33 (residual history), 34 (force history), 36 (Mach field), 37 (near-body pressure field), 38 (full-domain pressure field), 35 (wall C_p /skin-friction distribution).

9.6 NACA0012 M=2.0 laminar Re=5000

Status: **converged**. 5587 pseudo-time steps, residual reduction 3.27 orders (final L2 5.340e-04). Final force coefficients $C_D = 0.10132$, $C_L = 0.00025$, $C_{m_z} = -0.00010$; the final force row corresponds to the final field/surface state. The evidence figures for this case are 45 (residual history), 46 (force history), 48 (Mach field), 49 (near-body pressure field), 50 (full-domain pressure field), 47 (wall C_p /skin-friction distribution).

9.7 cylinder M=0.1 laminar Re=20

Status: **converged**. 969 pseudo-time steps, residual reduction 6.97 orders (final L2 1.077e-07). Final force coefficients $C_D = 2.10341$, $C_L = 0.00554$, $C_{m_z} = 0.00103$; the final force row corresponds to the final field/surface state. The evidence figures for this case are 1 (residual history), 2 (force

history), 4 (Mach field), 5 (near-body pressure field), 6 (full-domain pressure field), 3 (wall C_p /skin-friction distribution), 7 (vorticity field).

9.8 cylinder M=0.1 laminar Re=200

Status: **statistically_periodic**. Final time $t_f = 300.0$ after 30000 physical steps; final $C_D = 1.1461$, $C_L = 0.4438$. Post-startup mean $\overline{C_D} = 1.2322$, lift amplitude $\Delta C_L = 2.1238$ (min -1.0615 / max 1.0623). The dominant shedding frequency gives $St = 0.211$. The evidence figures for this case are 8 (residual history), 9 (force history), 11 (Mach field), 12 (near-body pressure field), 13 (full-domain pressure field), 10 (wall C_p /skin-friction distribution), 14 (vorticity field).

Figures: cylinder_m010_laminar_re20

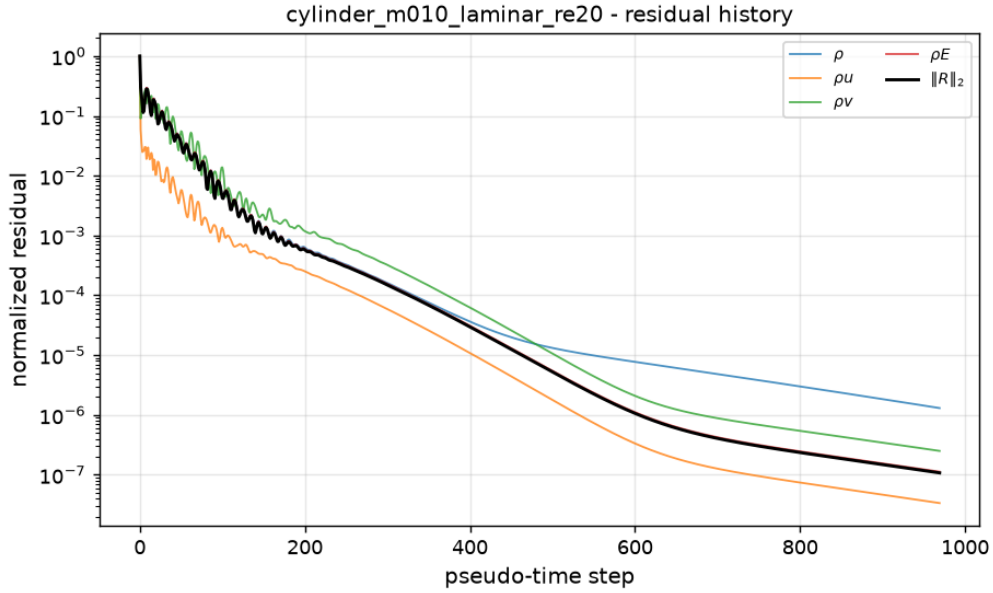


Figure 1: cylinder_m010_laminar_re20: global L2 residual and component residuals versus step (log scale).

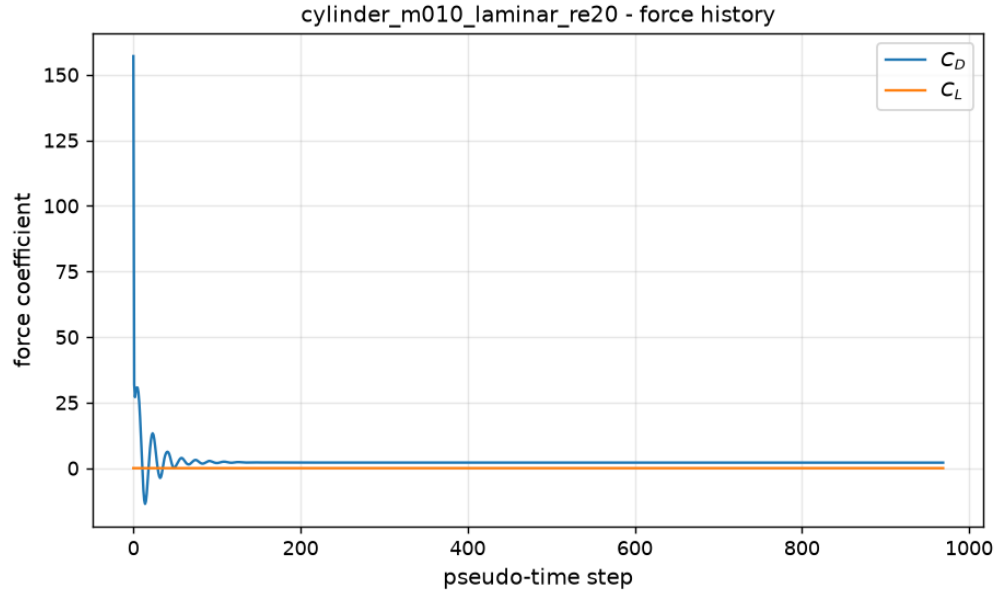


Figure 2: cylinder_m010_laminar_re20: drag and lift coefficient history versus time/step.

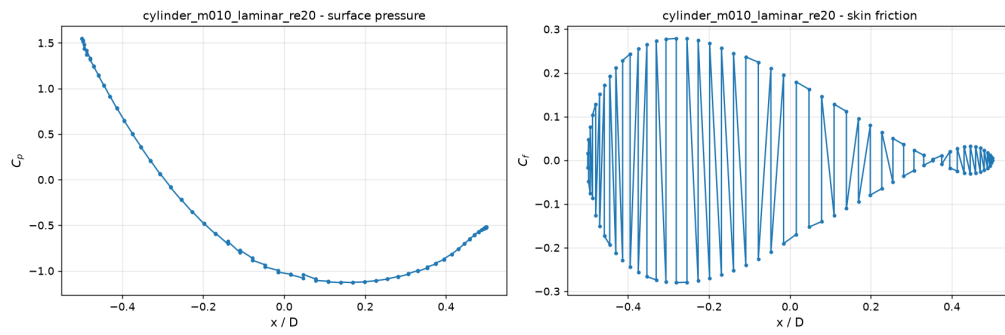


Figure 3: cylinder_m010_laminar_re20: wall pressure and skin-friction coefficient distributions.

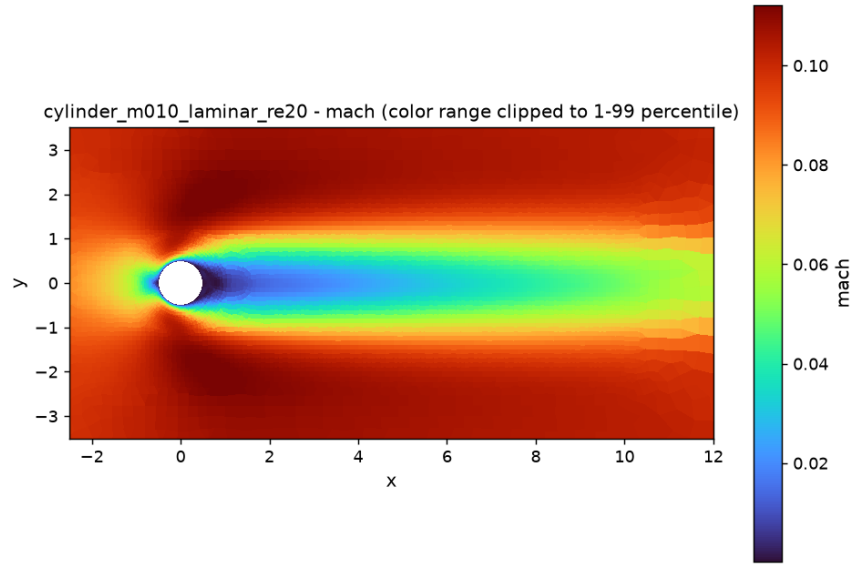


Figure 4: cylinder_m010_laminar_re20: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

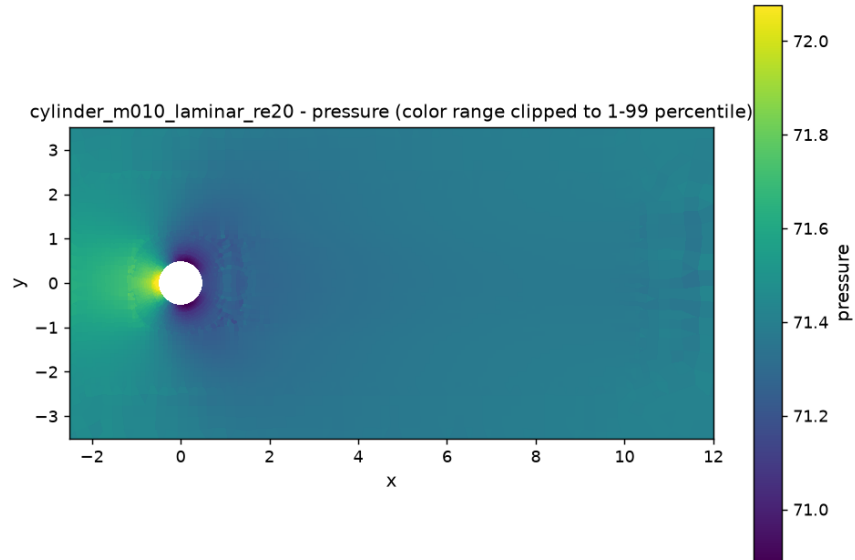


Figure 5: cylinder_m010_laminar_re20: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

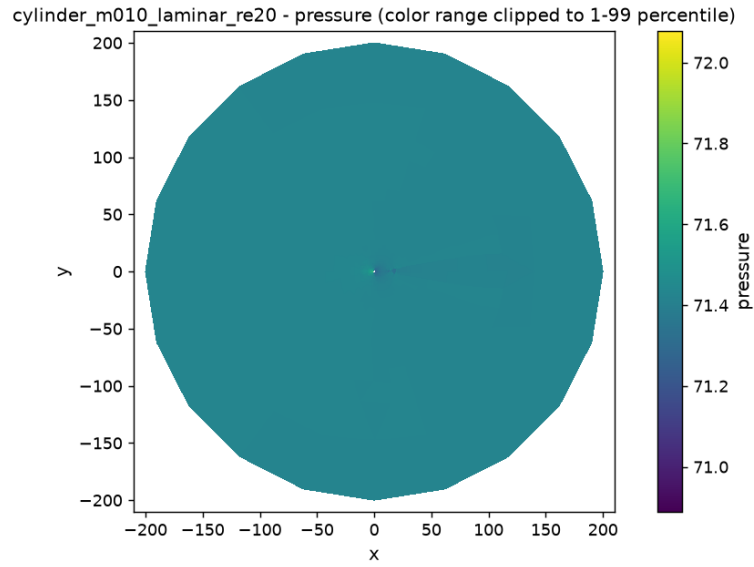


Figure 6: cylinder_m010_laminar_re20: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

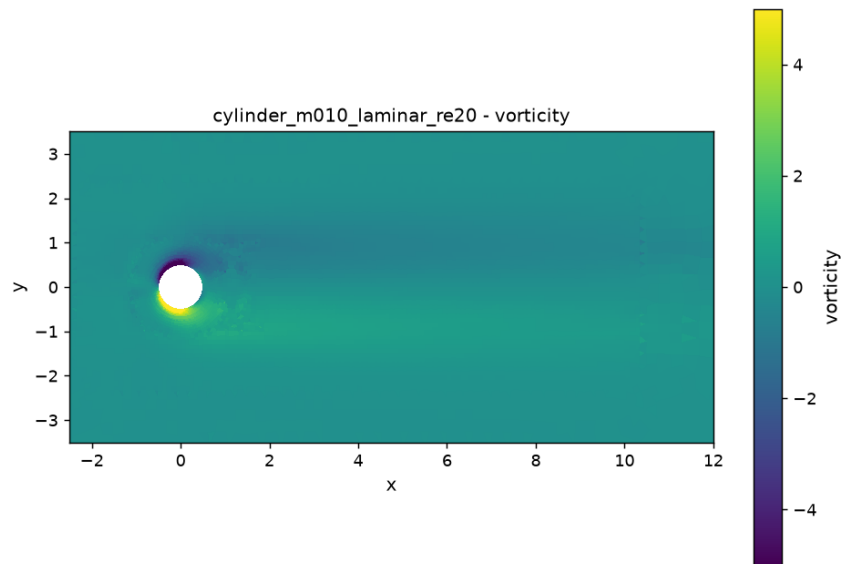


Figure 7: cylinder_m010_laminar_re20: vorticity field; color range clipped to $[-5, 5]$.

Figures: cylinder_m010_laminar_re200

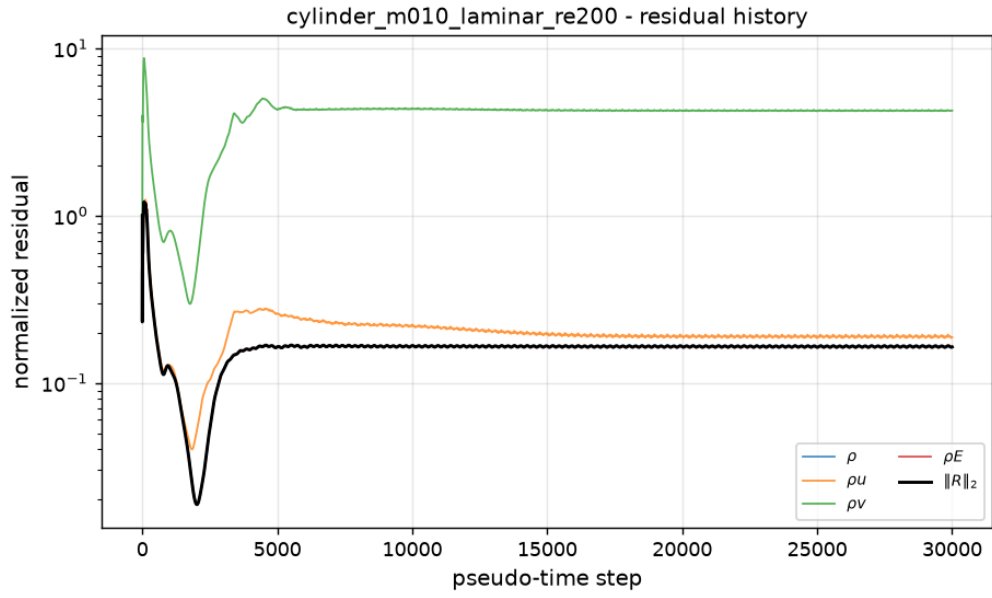


Figure 8: cylinder_m010_laminar_re200: global L2 residual and component residuals versus step (log scale).

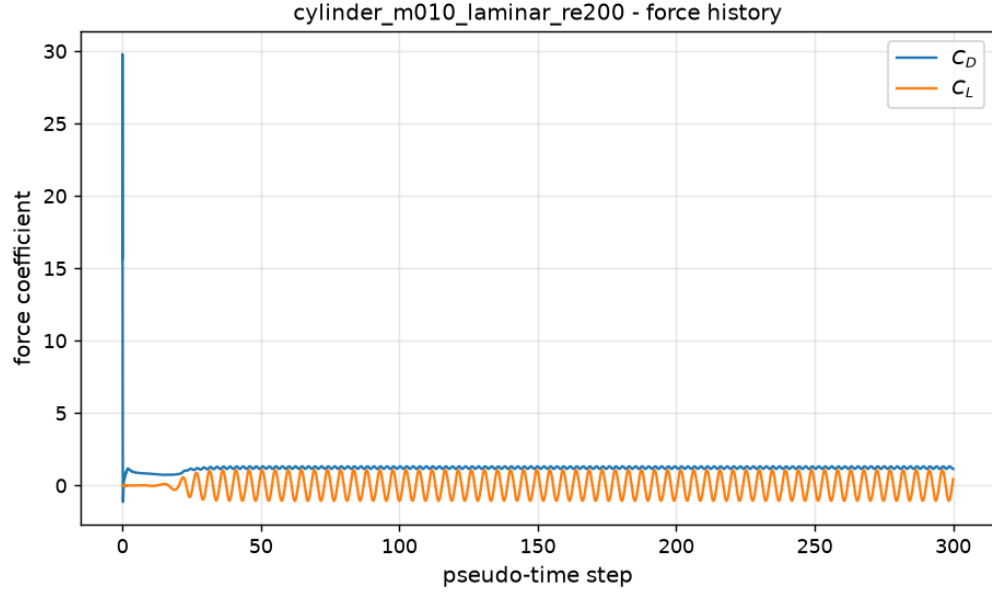


Figure 9: cylinder_m010_laminar_re200: drag and lift coefficient history versus time/step.

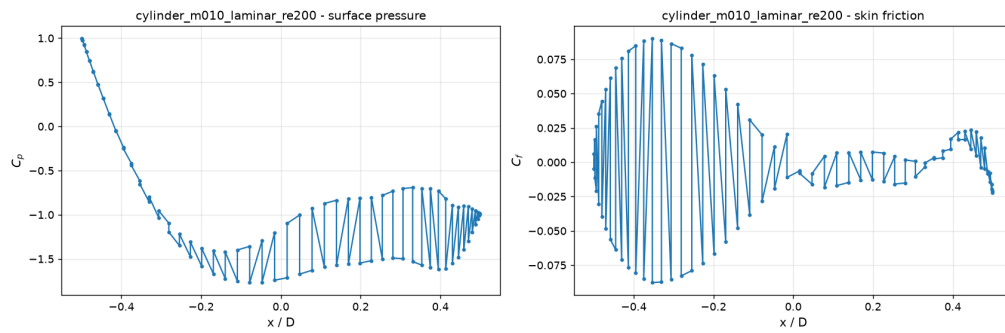


Figure 10: cylinder_m010_laminar_re200: wall pressure and skin-friction coefficient distributions.

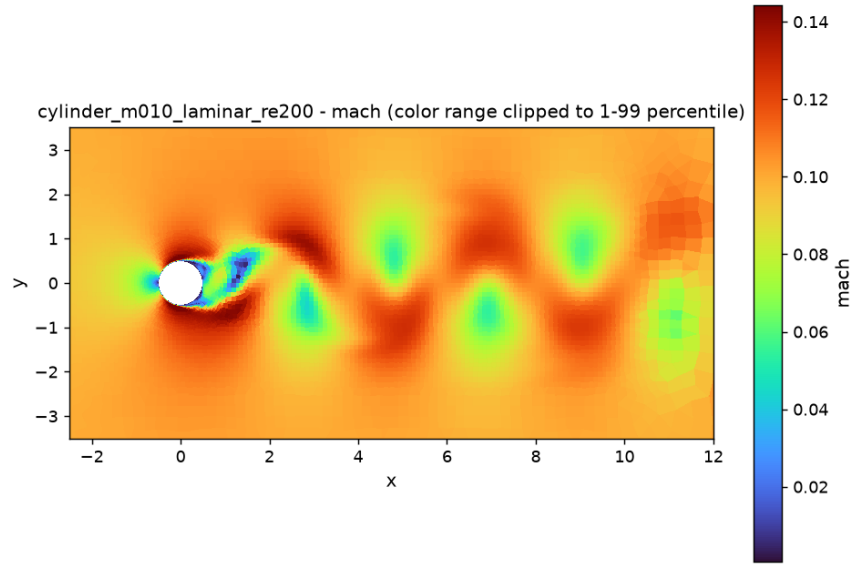


Figure 11: cylinder_m010_laminar_re200: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

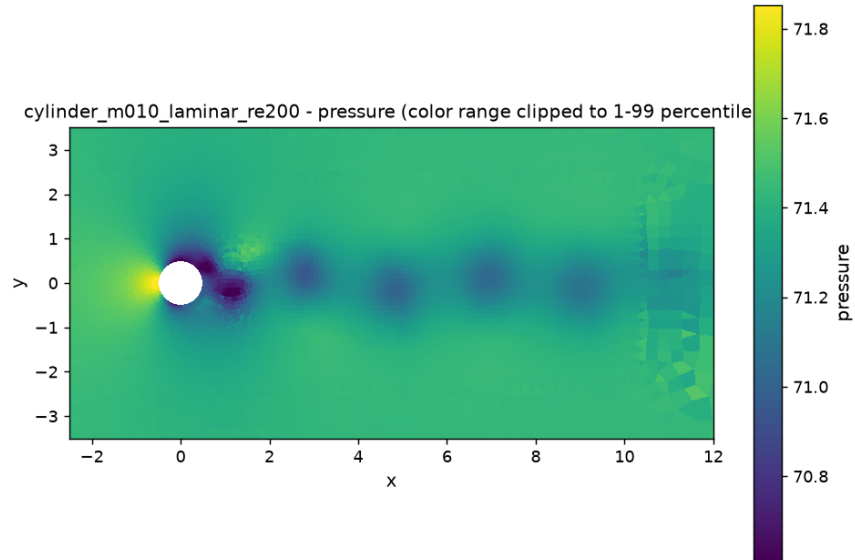


Figure 12: cylinder_m010_laminar_re200: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

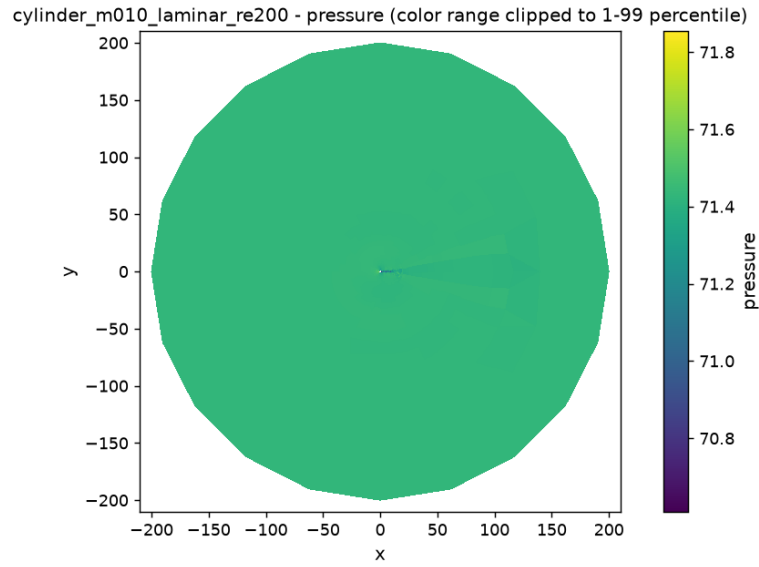


Figure 13: cylinder_m010_laminar_re200: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

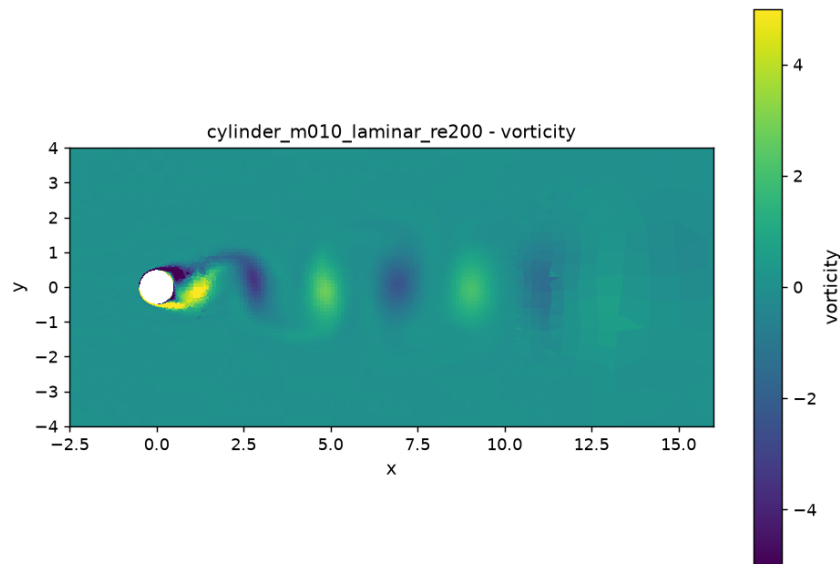


Figure 14: cylinder_m010_laminar_re200: vorticity field; color range clipped to $[-5, 5]$.

Figures: naca0012_m015_inviscid

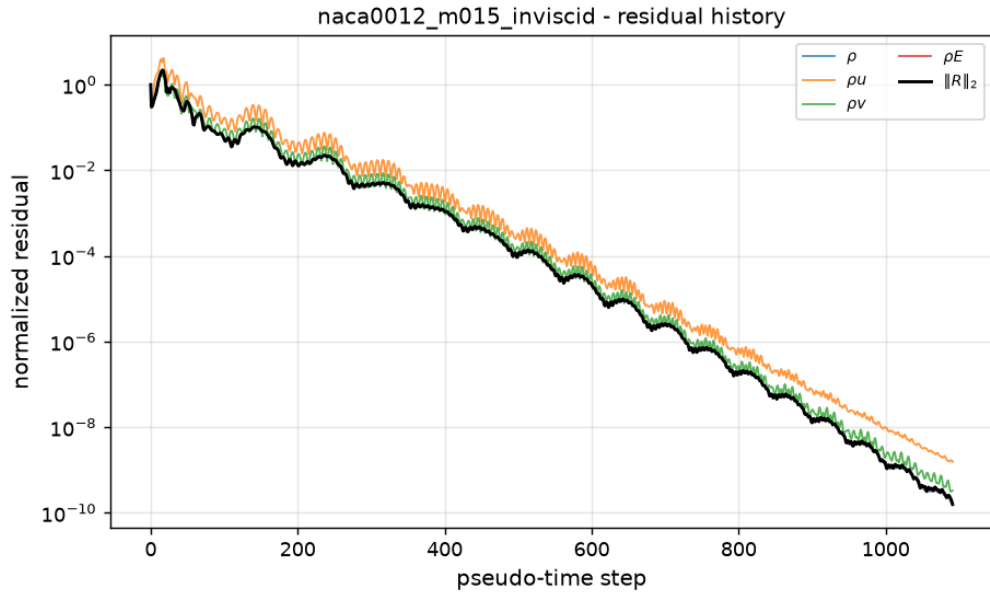


Figure 15: naca0012_m015_inviscid: global L2 residual and component residuals versus step (log scale).

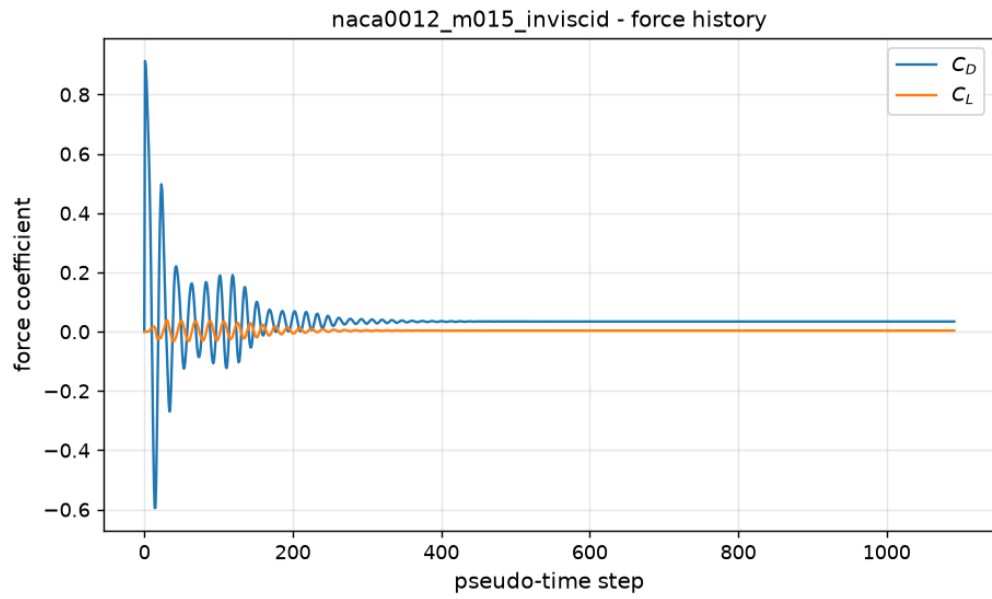


Figure 16: naca0012_m015_inviscid: drag and lift coefficient history versus time/step.

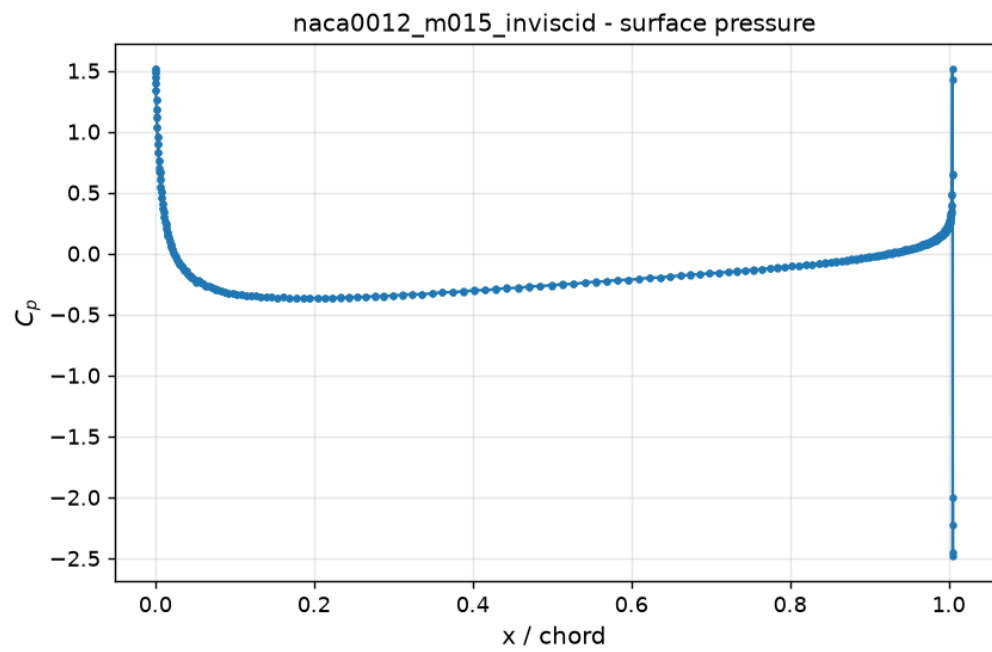


Figure 17: naca0012_m015_inviscid: wall pressure and skin-friction coefficient distributions.

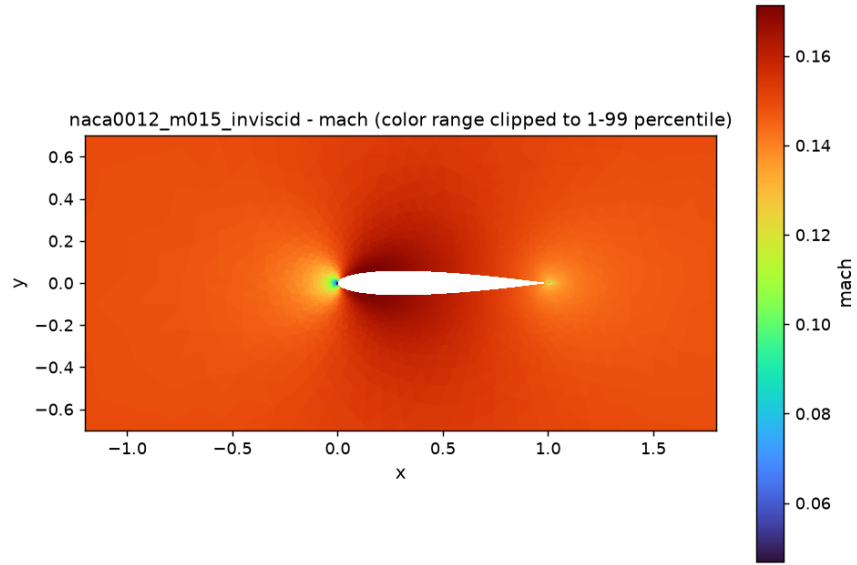


Figure 18: naca0012_m015_inviscid: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

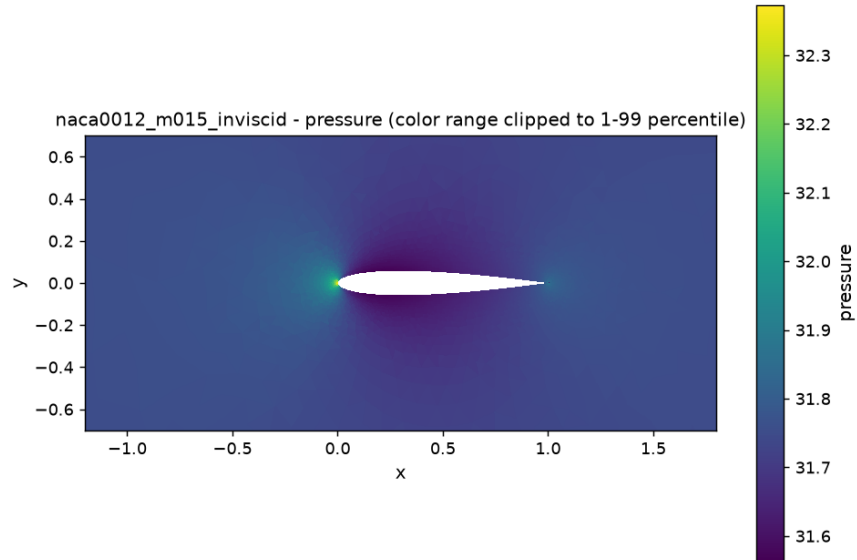


Figure 19: naca0012_m015_inviscid: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

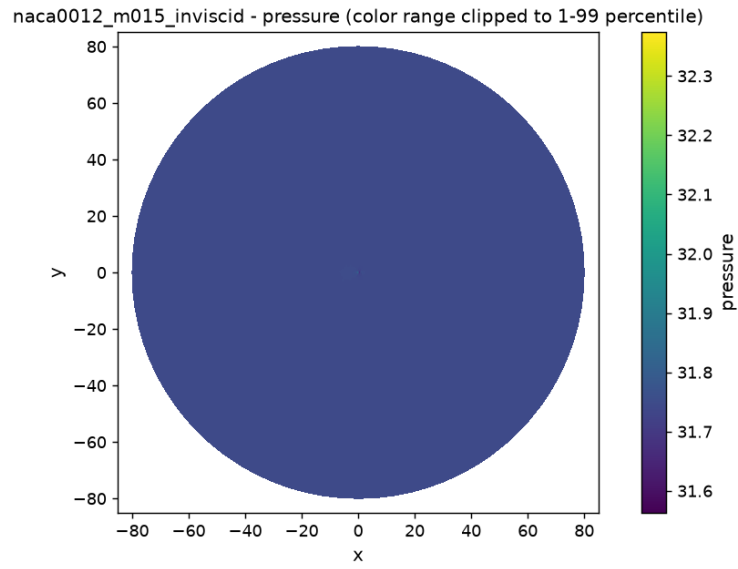


Figure 20: naca0012_m015_inviscid: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

Figures: naca0012_m015_laminar_re5000

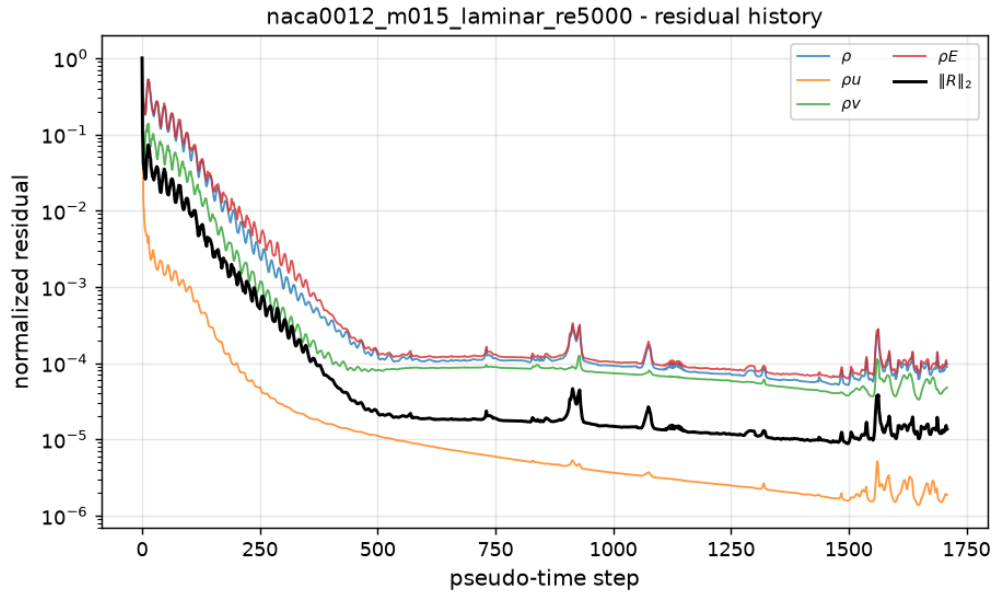


Figure 21: naca0012_m015_laminar_re5000: global L2 residual and component residuals versus step (log scale).

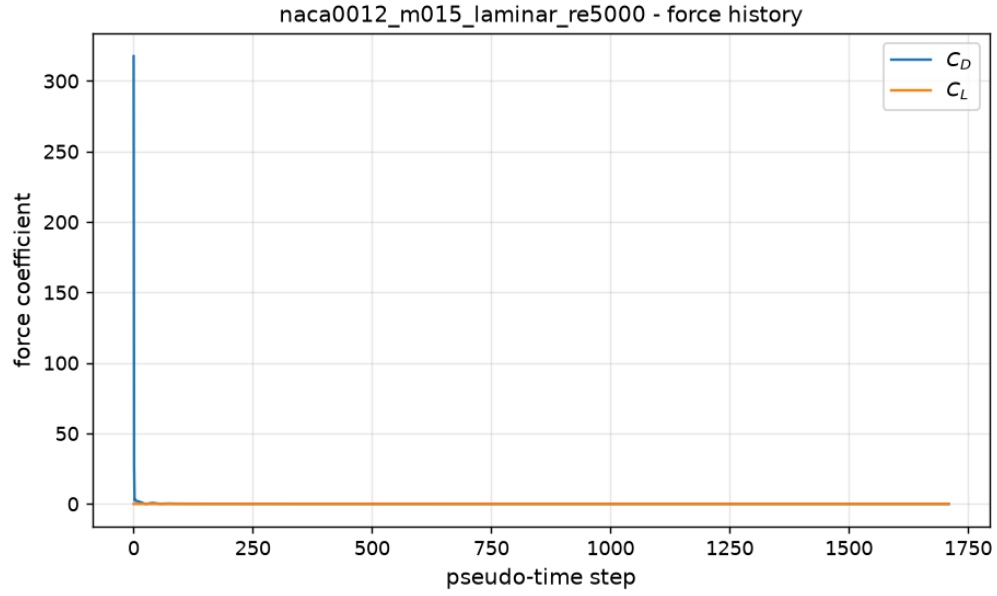


Figure 22: naca0012_m015_laminar_re5000: drag and lift coefficient history versus time/step.

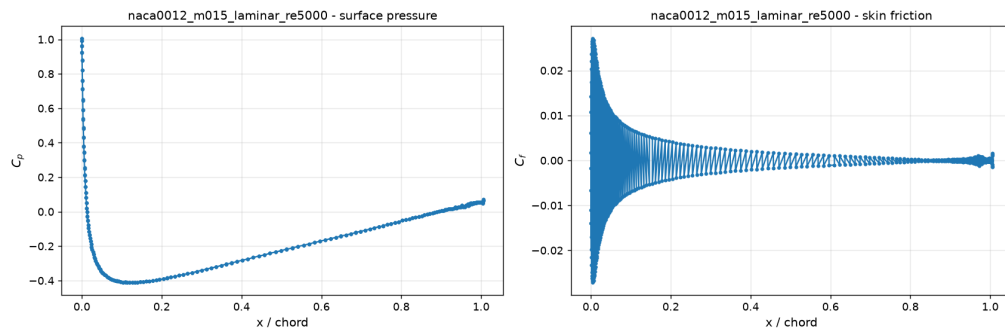


Figure 23: naca0012_m015_laminar_re5000: wall pressure and skin-friction coefficient distributions.

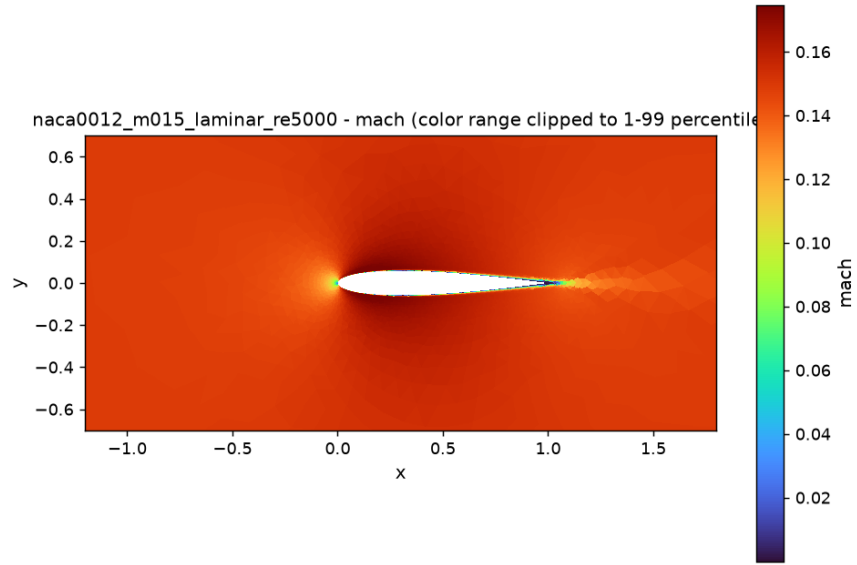


Figure 24: naca0012_m015_laminar_re5000: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

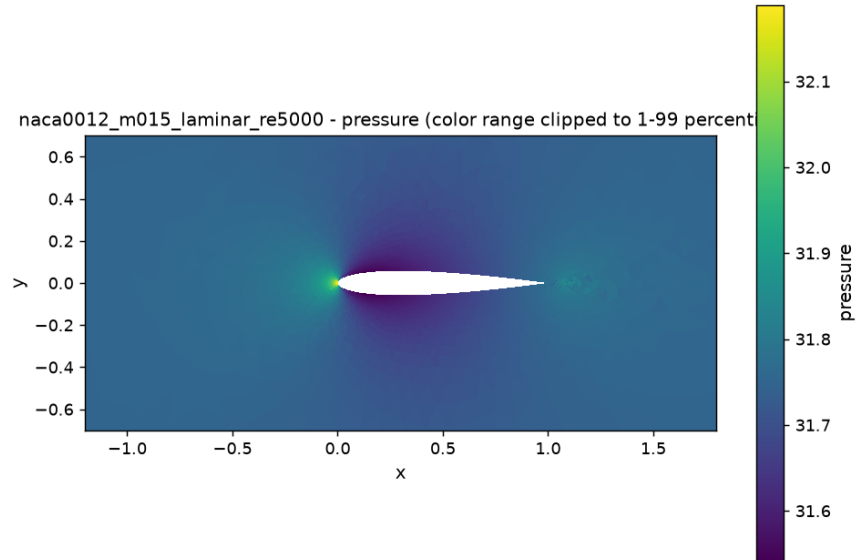


Figure 25: naca0012_m015_laminar_re5000: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

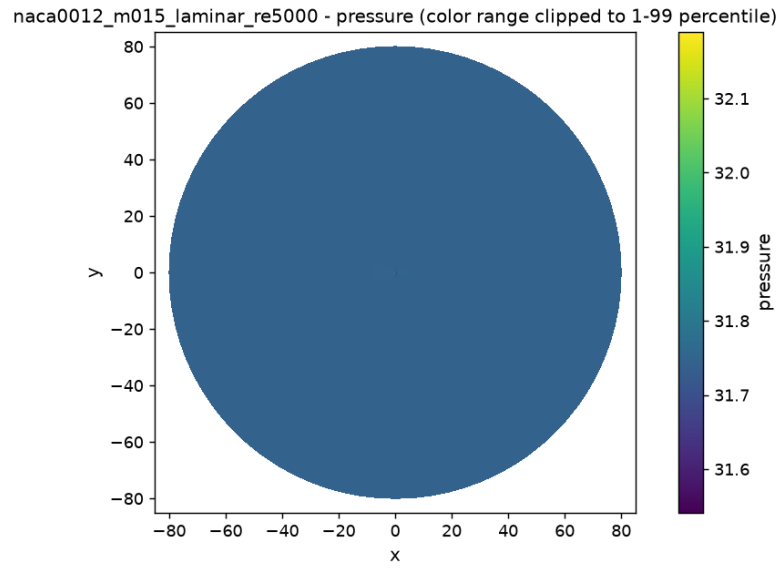


Figure 26: naca0012_m015_laminar_re5000: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

Figures: naca0012_m080_inviscid

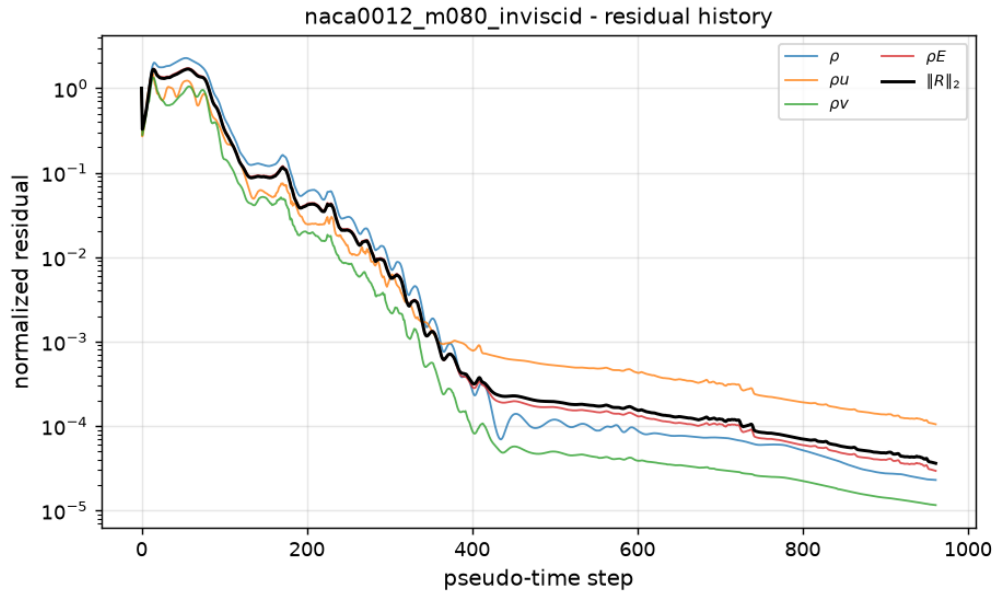


Figure 27: naca0012_m080_inviscid: global L2 residual and component residuals versus step (log scale).

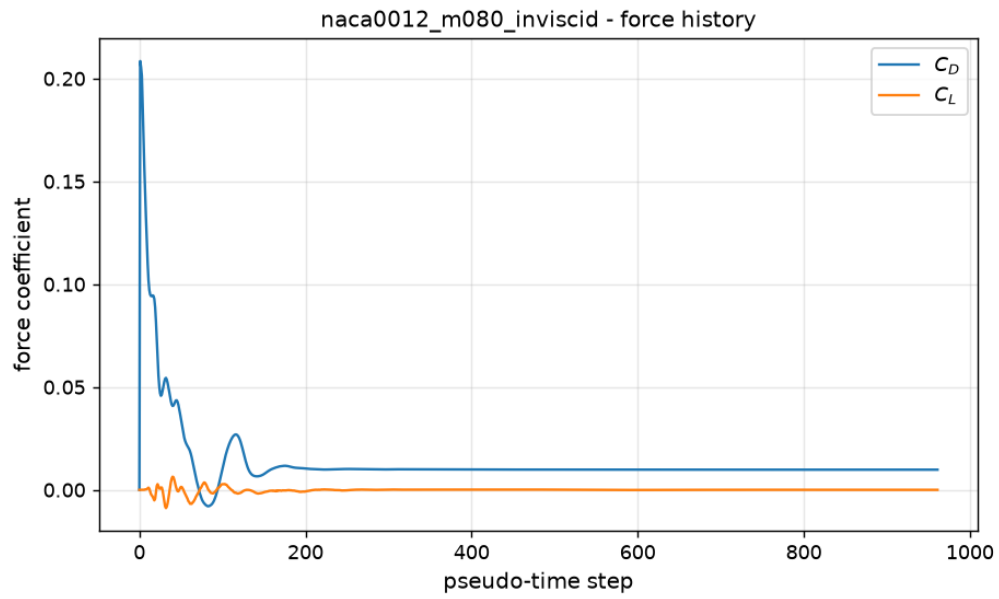


Figure 28: naca0012_m080_inviscid: drag and lift coefficient history versus time/step.

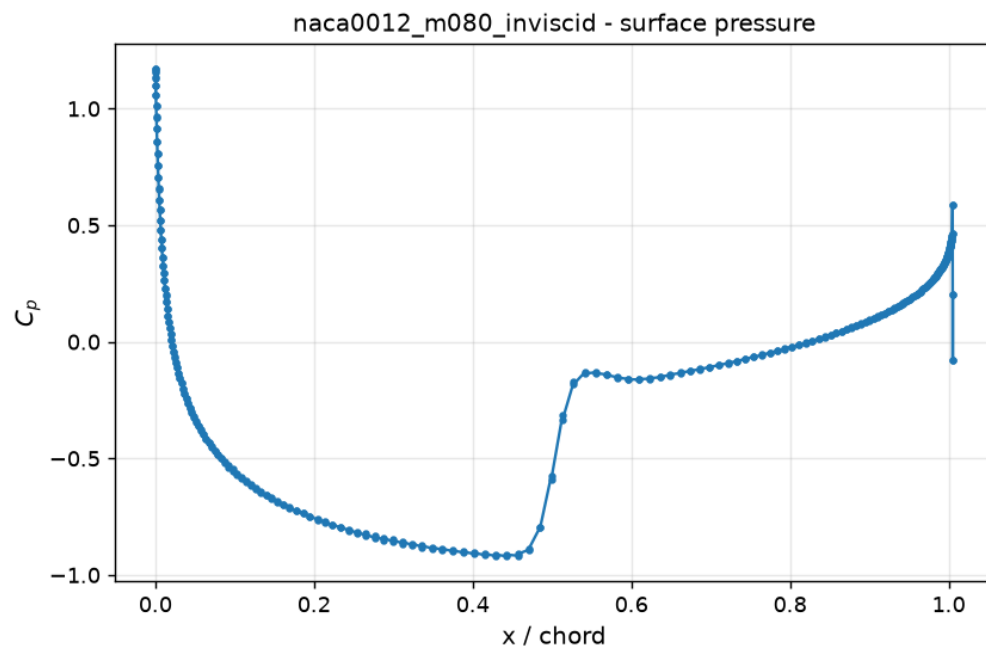


Figure 29: naca0012_m080_inviscid: wall pressure and skin-friction coefficient distributions.

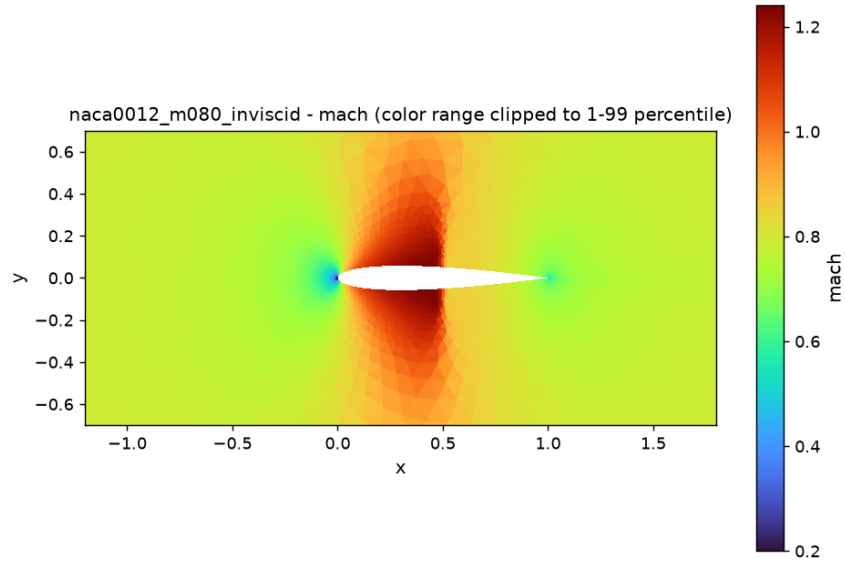


Figure 30: naca0012_m080_inviscid: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

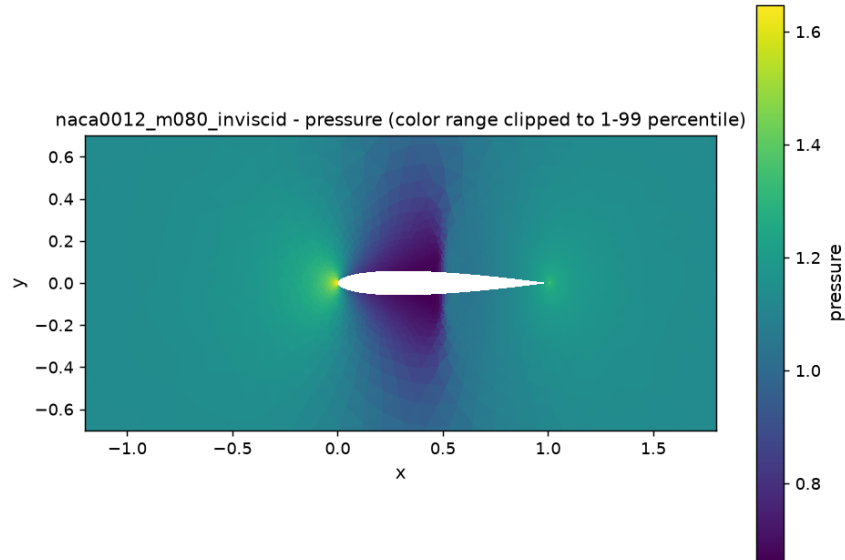


Figure 31: naca0012_m080_inviscid: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

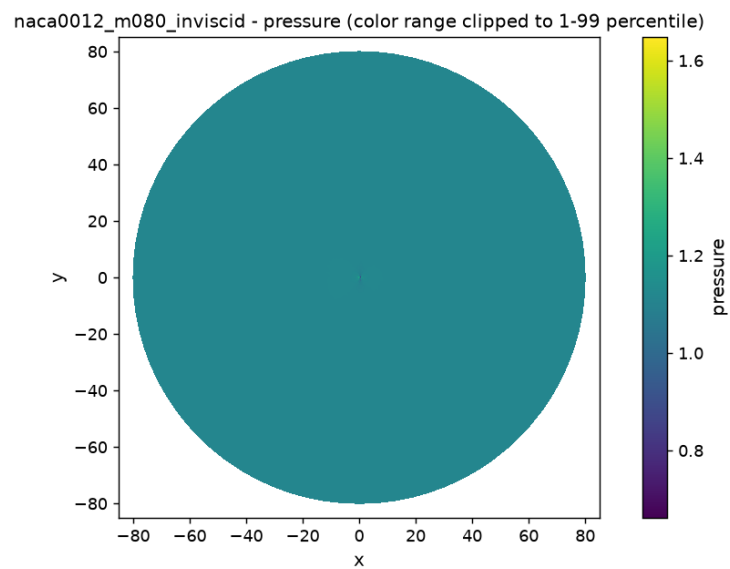


Figure 32: naca0012_m080_inviscid: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

Figures: naca0012_m080_laminar_re5000

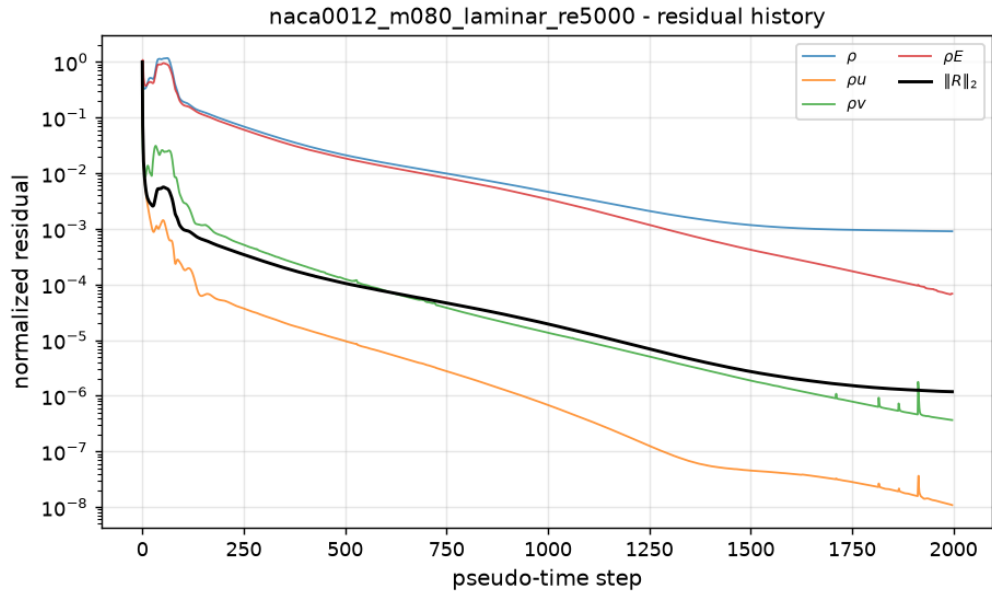


Figure 33: naca0012_m080_laminar_re5000: global L2 residual and component residuals versus step (log scale).

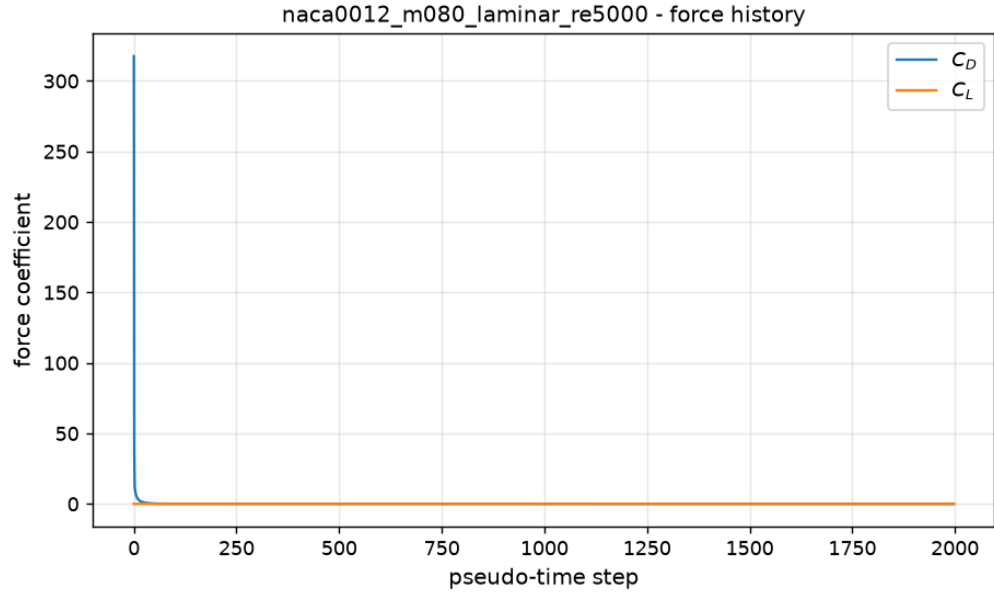


Figure 34: naca0012_m080_laminar_re5000: drag and lift coefficient history versus time/step.

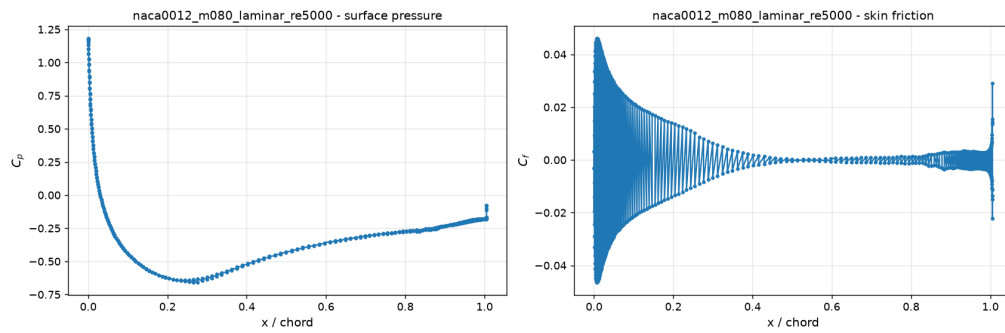


Figure 35: naca0012_m080_laminar_re5000: wall pressure and skin-friction coefficient distributions.

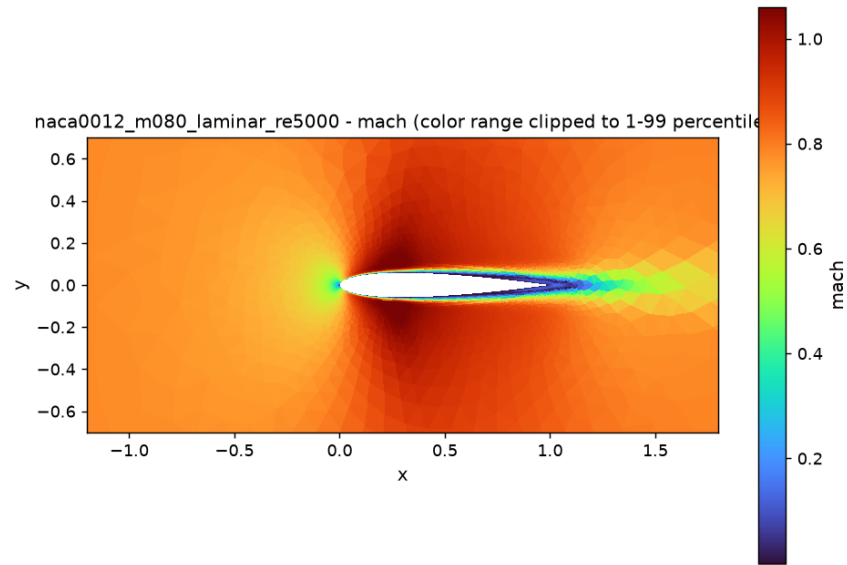


Figure 36: naca0012_m080_laminar_re5000: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

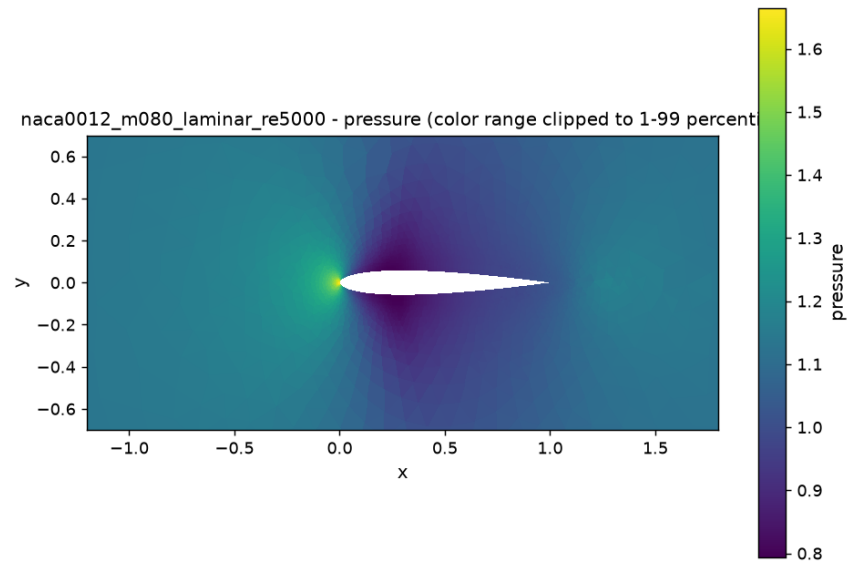


Figure 37: naca0012_m080_laminar_re5000: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

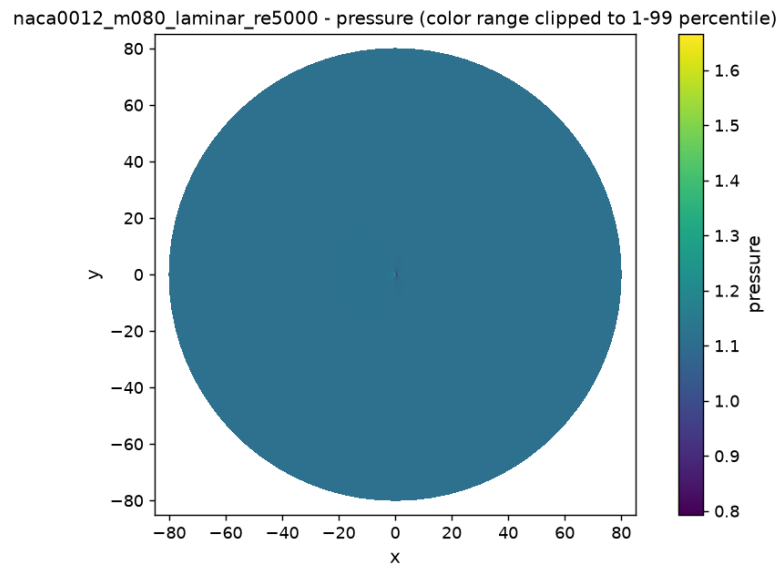


Figure 38: naca0012_m080_laminar_re5000: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

Figures: naca0012_m200_inviscid

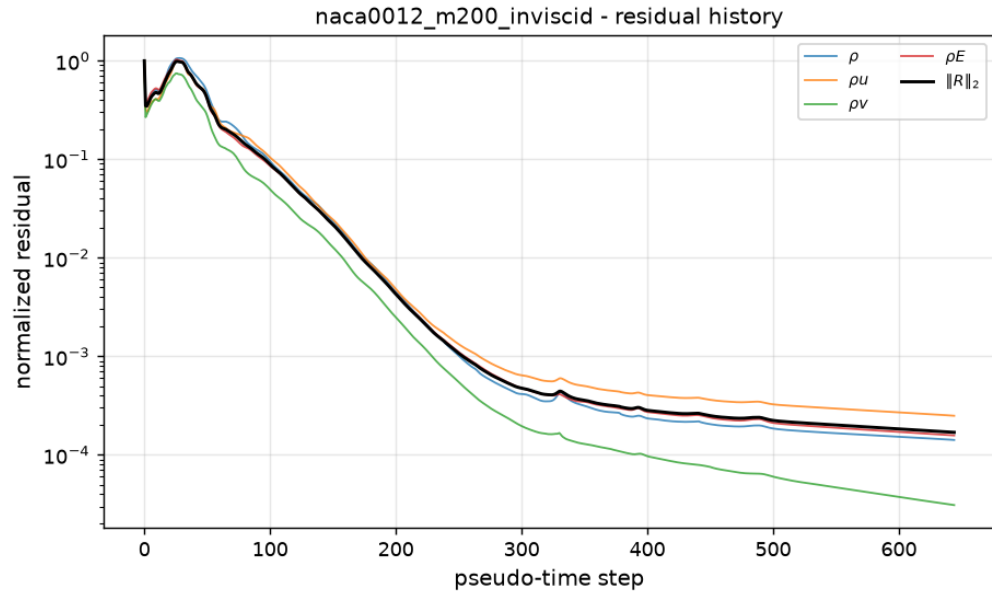


Figure 39: naca0012_m200_inviscid: global L2 residual and component residuals versus step (log scale).

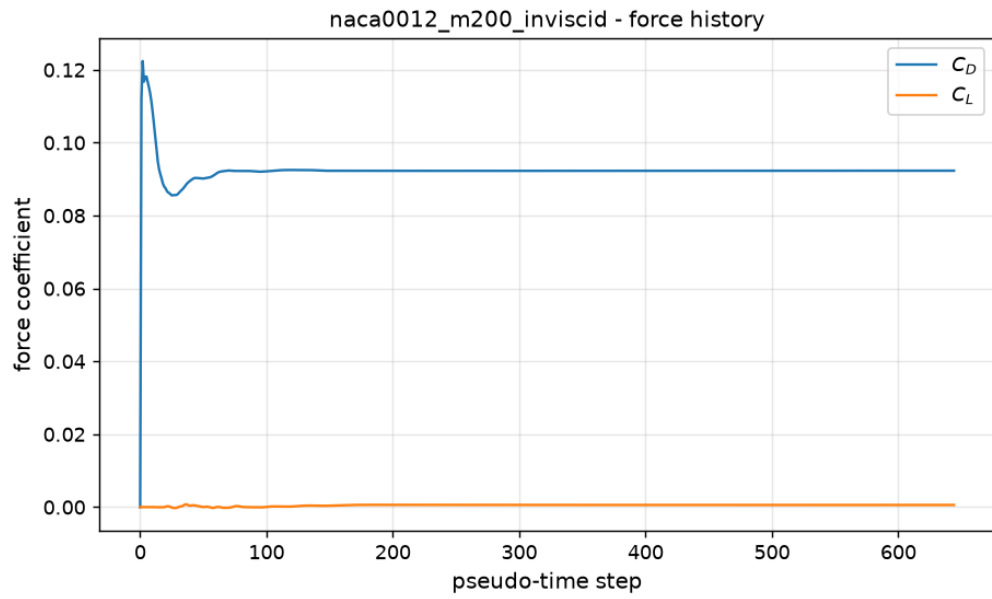


Figure 40: naca0012_m200_inviscid: drag and lift coefficient history versus time/step.

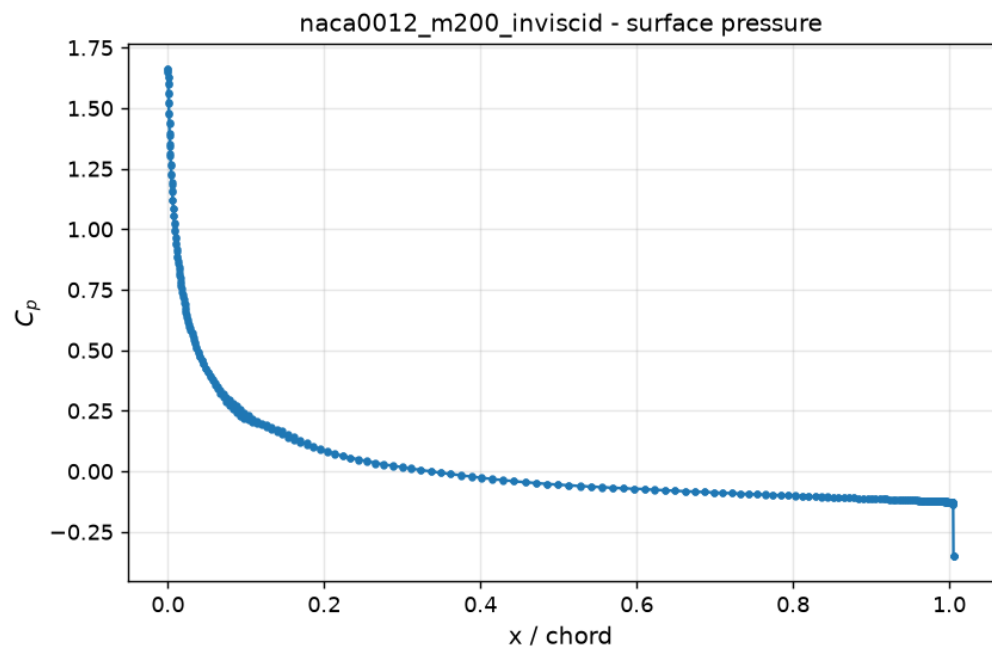


Figure 41: naca0012_m200_inviscid: wall pressure and skin-friction coefficient distributions.

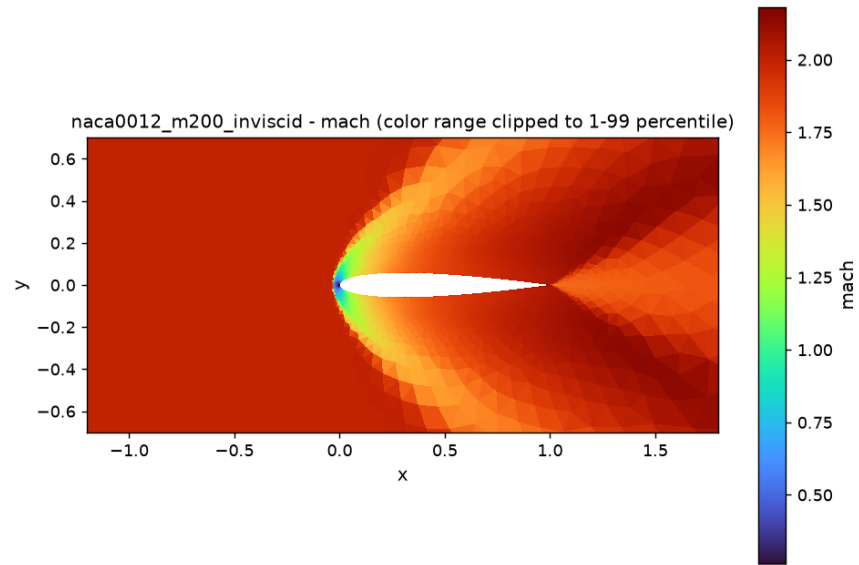


Figure 42: naca0012_m200_inviscid: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

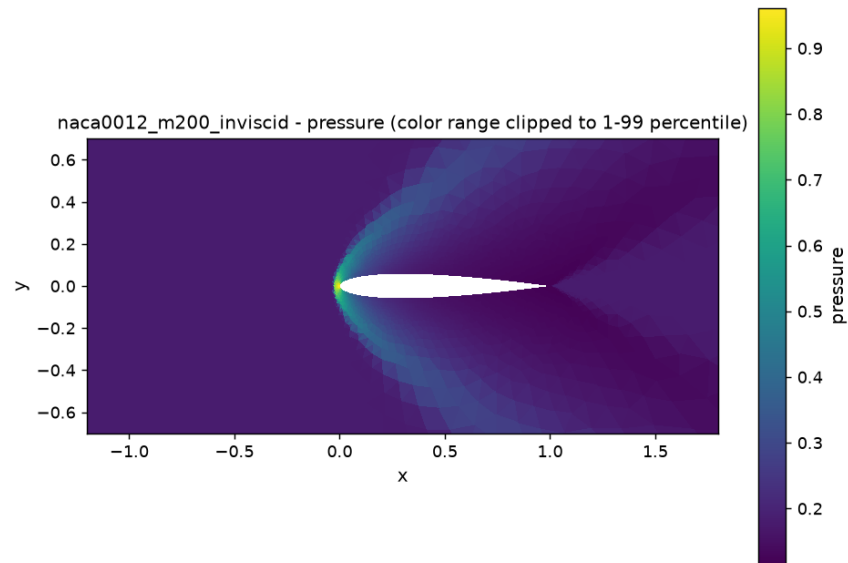


Figure 43: naca0012_m200_inviscid: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

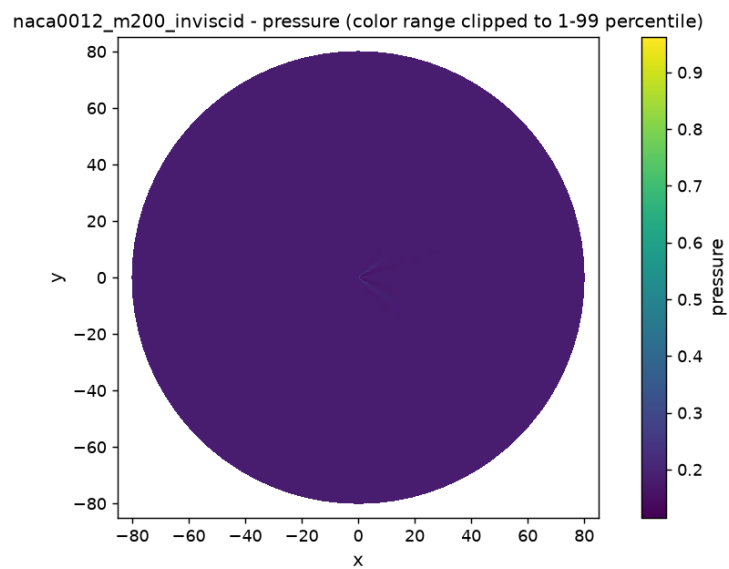


Figure 44: naca0012_m200_inviscid: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

Figures: naca0012_m200_laminar_re5000

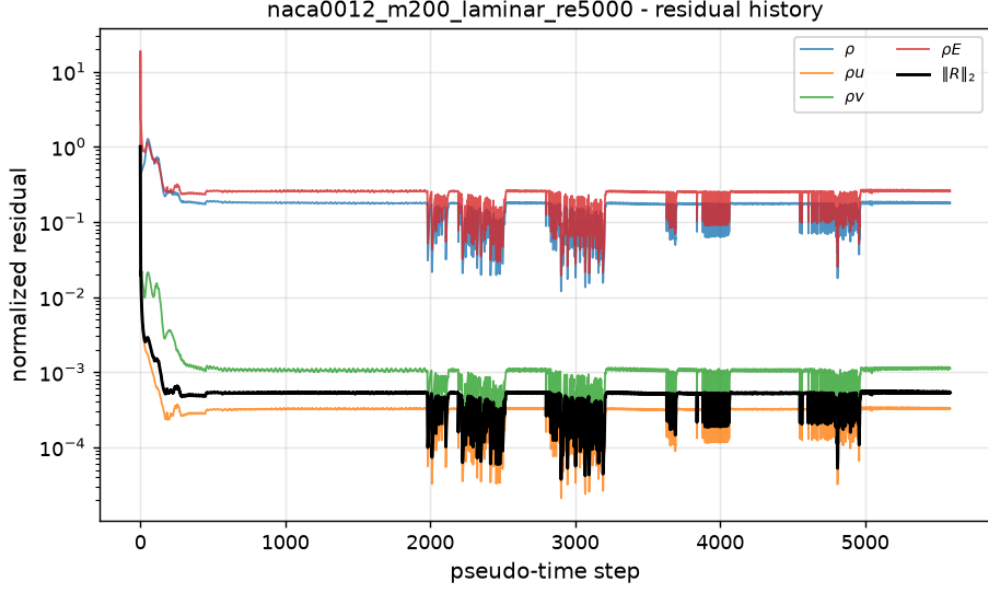


Figure 45: naca0012_m200_laminar_re5000: global L2 residual and component residuals versus step (log scale).

10 Parallel Validation

The NACA0012 $M=0.15$ inviscid case and the cylinder $Re=20$ case were each run to the same production convergence criteria at $np=1$ and $np=8$ (Table 1 reports the $np=8$ production wall times). Because every force row is assembled from an `MPI_Allreduce` over the rank-local wall-face contributions, the force integrations themselves are rank-count safe, so the comparison quantifies how much the converged state can shift when the parallel iteration path changes. Table 4 reports the result: the converged $np=1$ and $np=8$ coefficients agree to $|\Delta C_D| = 3.5 \times 10^{-4}$ for the NACA case and 2.3×10^{-2} (about 1.1% relative) for the cylinder case. Once the impulsive-start transient has decayed, the two force histories track each other at matched pseudo-time steps to those same levels, while the first few hundred ramp steps differ by more because the startup transient amplifies small iteration-path differences. The larger cylinder residue is the marginal trailing-edge/wake mode that `CFD_RECON_SCALE=0.5` damps but does not remove; its residual amplitude sets the rank-count reproducibility floor of that case.

The $np=8$ partition diagnostics for the two comparison cases are given in Table 3 (owned/ghost cells per rank, neighbor counts, and send/receive volumes); the edge cut, min/max/mean owned cells, and load-balance ratio summaries are in each case’s `partition_summary.json`. The observed imbalance is under 2% for both meshes, consistent with the METIS k-way objective and the serial-preprocessing/rank-local-solve design.

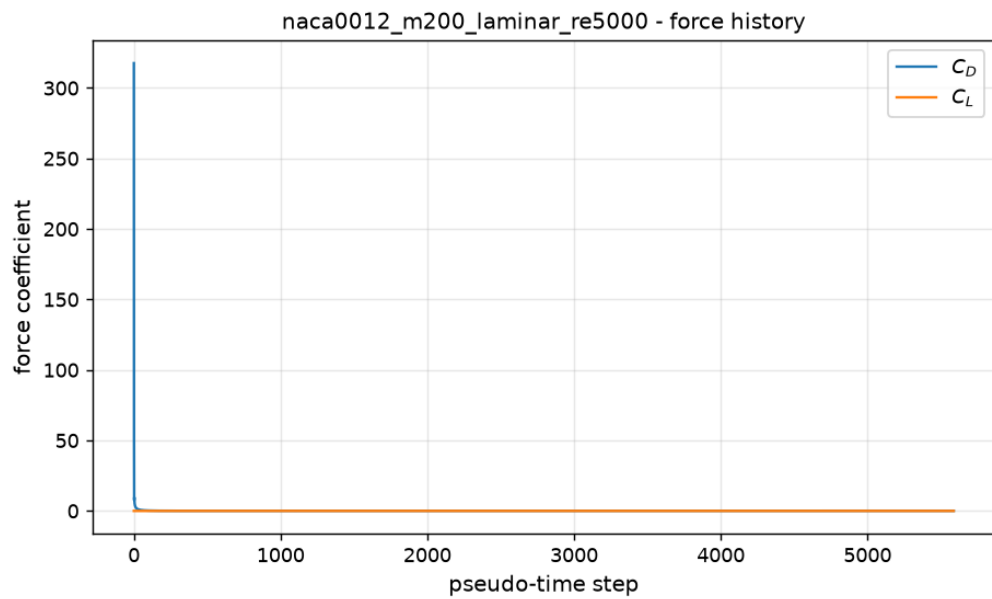


Figure 46: naca0012_m200_laminar_re5000: drag and lift coefficient history versus time/step.

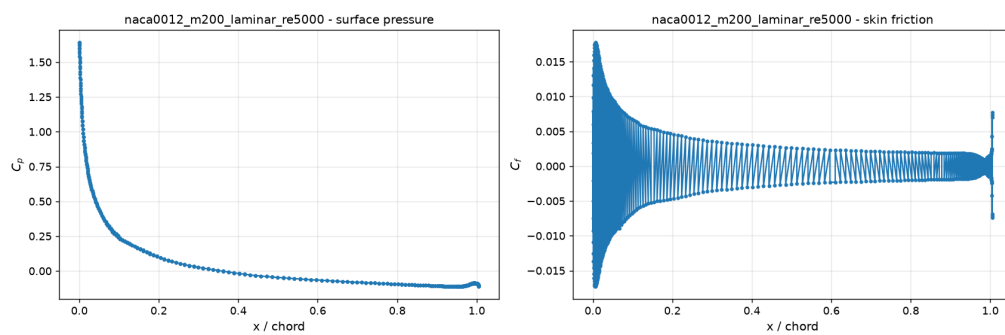


Figure 47: naca0012_m200_laminar_re5000: wall pressure and skin-friction coefficient distributions.

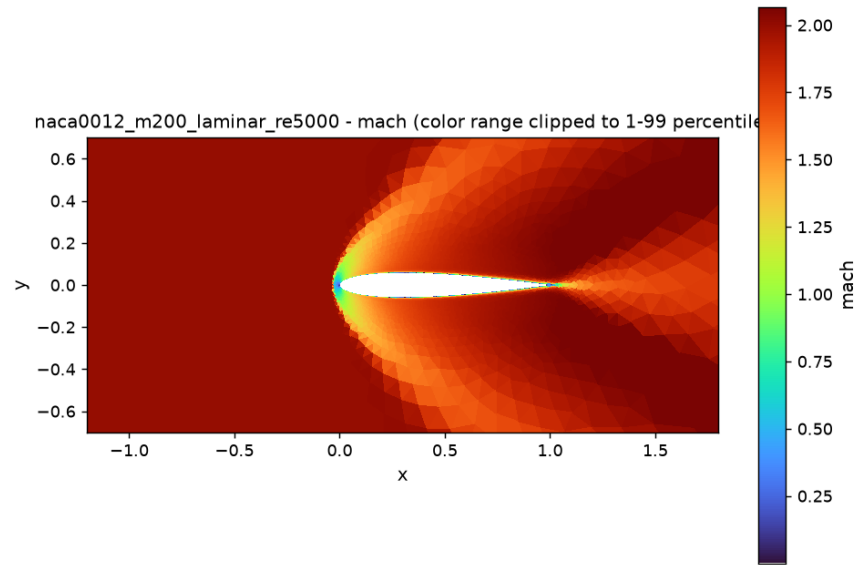


Figure 48: naca0012_m200_laminar_re5000: Mach number field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

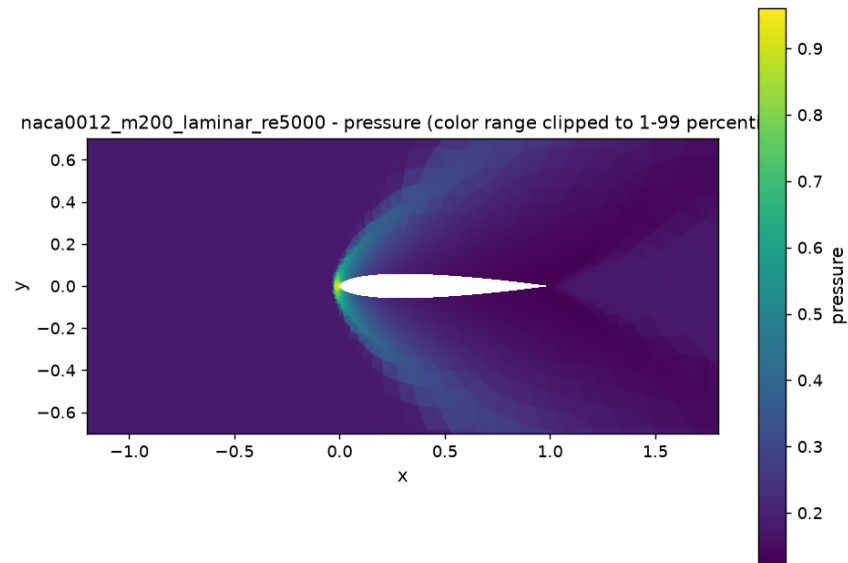


Figure 49: naca0012_m200_laminar_re5000: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

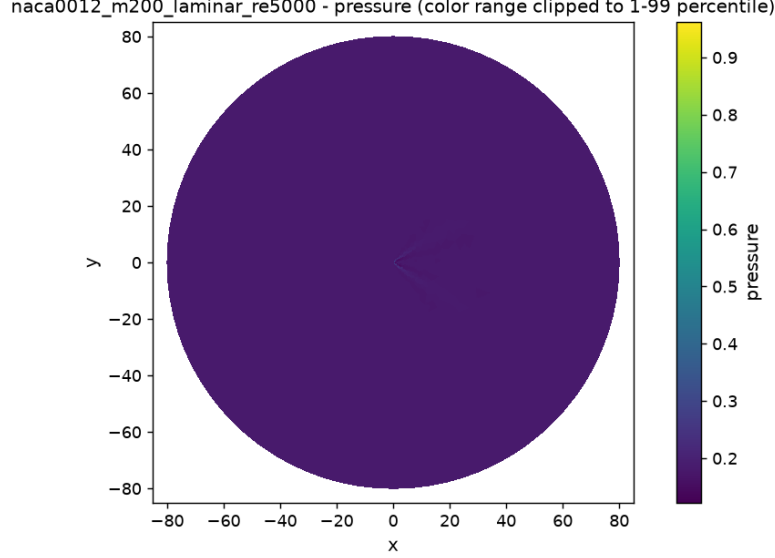


Figure 50: naca0012_m200_laminar_re5000: nondimensional pressure field rendered on the unstructured mesh, color range clipped to the 1-99 percentile.

11 Verification-Driven Debugging

Two genuine discretization bugs were found during verification and were fixed before the production results in this report were generated; the fix commits ([af9980a](#), [a151896](#)) are in the repository history.

Self-face neighbor lookup. The Green–Gauss gradient assembly and both limiter implementations originally indexed the face neighbor as `prim[face_right]` without consulting which side of the face the current cell occupies. On this mesh about half of every cell’s faces are “right” faces, so the neighbor was the cell itself for those faces. In the Venkatakrishnan limiter the self-difference 0 enters the neighbor min/max range, and in the cylinder wake this collapsed the u -reconstruction to a median limiter factor $\phi(u) = 0.33$ (55% of wake cells below 0.5). The scheme therefore behaved first-order exactly where the vortex-street mode lives, adding a numerical viscosity $\nu_{\text{num}} \sim 0.5|u|\Delta x \approx 0.01\text{--}0.05$ against the physical $\nu = 0.005$, i.e. an effective Reynolds number near or below the shedding threshold $Re_c \approx 47$. Measured consequences: the seeded wake perturbation decayed at $-0.14 \dots -0.28$ per convective time instead of growing, and the pre-fix diagnostic runs converged to a steady attached wake with $C_D \approx 1.62$ and essentially frozen lift (no shedding); the retained `results/diag_*` artifacts and `PROGRESS.md` record this. The fix uses the true neighbor (`outward ? face_right : face_left`) in the gradient and both limiters, recovering $\phi(u) \approx 1$ in the wake.

Wall ghost-gradient anchor. The no-slip wall viscous flux and the wall force integration built their ghost-cell gradient correction from the *reconstructed* face state q_{Lf} . Once the corrected gradients make the reconstruction satisfy $u \approx 0$ at the wall face, the enforced jump $q_g - q_{Lf}$ vanished and with it the wall shear (the fixed-operator runs showed viscous drag ≈ -0.03 and near-zero total drag). The correction now anchors the jump to the cell-center state q_{Lc} , which is the state the ghost-to-cell-center distance ΔL_G actually spans.

12 Extensibility and Maintainability

The code is organized so that extending the solver does not require rewriting the solver loop. Case inputs are fully external: every physics setting, boundary-family mapping, CFL control, and stopping criterion comes from the case JSON (`src/config.cpp`); no case name, mesh name, or reference force value appears anywhere in the solver logic. The equation of state is isolated in the `GasModel` interface (`src/common.hpp`), which centralizes all perfect-gas closures (p , a , T , primitive/conservative conversion) behind the case-configurable γ , R , and Prandtl number, so a general EOS or multi-species model can replace one well-defined interface without touching the residual assembly. Boundary conditions are dispatched from the JSON family map through an extensible farfield/slip-wall/no-slip-adiabatic enum switch (`src/partition.cpp`, `src/solver.cpp`); adding a boundary type is additive. The mesh abstraction (`src/mesh.hpp`) stores faces with unit normals, areas, and left/right connectivity, and the residual is written in face-normal form, so it does not assume a specific cell topology.

The main places that are currently 2-D-specific are the `Vec2` and primitive-gradient types, the CGNS coordinate read, the Roe eigenvector expansion, the viscous stress/heat-flux terms, and the VTU writer. All of these are localized to their modules; a 3-D port is therefore a mechanical, per-module change rather than a redesign. Turbulence modeling and species transport are not implemented: the code solves the single-species laminar Navier–Stokes equations with a scalar viscosity, and the seams for RANS (a turbulence model feeding the effective μ/k in `src/numerics.cpp`) and for species (a generalized `GasModel/PrimState`) are documented here and in the README. The source is split into `config/cgns_reader/mesh/partition/numerics/solver/io` units with checked JSON parsing, bounded per-rank memory, and no monolithic scripts in the solve path.

13 Limitations and Failure Analysis

The supersonic and inviscid steady cases use the Rusanov (local Lax–Friedrichs) inviscid flux, which is stable and adequate there because the flow speed is comparable to the acoustic speed. At the low freestream Mach numbers of the cylinder cases ($M_\infty = 0.1$) Rusanov dissipation scales with the acoustic speed and acts like an artificial viscosity roughly two orders of magnitude above the physical μ : an initial 30,000-step Re 200 Rusanov production run converged to a steady attached wake with no recirculation and no shedding, and the Re 20 run showed the same inflated, bubble-free wake with $C_D \approx 4.05$ instead of the expected ≈ 2.0 (retained in `results/old_rusanov_re20_partial`). Those low-Mach viscous cases (cylinder Re 20 and Re 200, NACA M=0.15/0.8 laminar) were therefore re-run with the Roe/Harten–Hyman flux, whose wave-appropriate dissipation restores the expected recirculating wakes and force levels; the supersonic and inviscid steady cases retain Rusanov, and the residual dissipation effect on the M=0.15 inviscid drag is reported honestly below. The simplified-Jacobian LU-SGS relaxation is a first-order-accurate linearization of the residual operator; it converges robustly in all production cases at the cost of using the full case-supplied inner-iteration budget on every step. The Venkatakrishnan limiter parameters ($K = 0.3$) and the positivity fallback were kept at their validated defaults rather than being re-tuned per case.

The Re 200 dual-time solve meets the inner residual target of 10^{-3} on every physical step (the case metadata records `inner_target_converged_fraction=1.0` and zero `inner_target_misses`); the first physical step uses backward Euler by construction, while all later steps use the established BDF2 three-point stencil. A small asymmetric initial perturbation (about 2% of the freestream speed in the near wake) is applied so the linearly unstable wake mode saturates into the periodic vortex street on the expected time scale instead of waiting for round-off asymmetry; it does not

affect the saturated state or the Strouhal number. The wall-clock time of the Re 200 production run reflects the matrix-free Newton–GMRES inner solve with one LU-SGS preconditioner sweep; a fully differentiated Jacobian would reduce inner iterations but was not required by the contract. No case diverged: all submitted runs finished with the convergence statuses reported in Table 1, and every numerical experiment that did fail (e.g. the early Barth-limited runs, the Rusanov Re 200 vortex-street attempt, and one diagnostic farfield variant) is documented in the repository history and excluded from the final results.